

# Multiparameter Full-Waveform Inversion and the Cross-Talk Problem

A self-contained technical tutorial: from the adjoint state and the Hessian to radiation patterns, parameterization, preconditioning and truncated-Newton methods

Prepared for the `tdfwi-hess` multiparameter elastic FWI project

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## Abstract

Full-waveform inversion (FWI) estimates subsurface properties by minimising the misfit between recorded and simulated seismograms. When more than one physical parameter is sought, for example the P- and S-wave velocities and density of an isotropic elastic medium, or velocity and Thomsen anisotropy parameters, the inversion must separate the imprint that each parameter leaves on the data. Imperfect separation, known as *cross-talk* or *inter-parameter trade-off*, is the central difficulty of multiparameter FWI. This tutorial develops, from first principles, everything needed to understand and attack this problem. We review least-squares inverse theory, the physics and parameterizations of acoustic, elastic and anisotropic media, and the adjoint-state computation of gradients. We then derive the Gauss–Newton and full Hessians and the second-order adjoint and finite-difference Hessian-vector products. Born scattering theory gives the radiation patterns that are the local columns of the sensitivity operator, and we show analytically and numerically that cross-talk is the off-diagonal structure of the multiparameter Hessian. That structure is controlled by the overlap of radiation patterns over the available scattering-angle aperture. We then survey and compare remedies: parameterization choice, scaling, diagonal and block-diagonal preconditioning, quasi-Newton and truncated-Newton optimisation, hierarchical and data-selective strategies, regularisation and prior information, multicomponent data, and resolution analysis. Every quantitative claim is reproduced by a short accompanying script, `make_figures.py`, which also provides a numerical laboratory: a linearised elastic amplitude-versus-angle inversion in which the cross-talk of the gradient, its partial correction by diagonal scaling, its local removal by  $3 \times 3$  block preconditioning and its global removal by Gauss–Newton can be measured directly. The paper closes with implementation guidance (verification tests, finite-difference step selection in single precision, cost accounting) and maps the theory onto the matrix-free Newton-CG implementation in `tdfwi-hess`.

## Contents

|          |                                                    |          |
|----------|----------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                | <b>4</b> |
| 1.1      | Why multiparameter FWI?                            | 4        |
| 1.2      | The cross-talk problem in one paragraph            | 4        |
| 1.3      | Three faces of cross-talk                          | 4        |
| 1.4      | What this tutorial contains                        | 5        |
| 1.5      | Notation                                           | 5        |
| <b>2</b> | <b>Foundations: least-squares inverse problems</b> | <b>6</b> |
| 2.1      | The forward problem as an operator                 | 6        |

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| 2.2      | Misfit functions and their statistical meaning . . . . .              | 6         |
| 2.3      | Linearisation: Fréchet derivative and adjoint . . . . .               | 6         |
| 2.4      | Second derivatives and the Newton equation . . . . .                  | 7         |
| 2.5      | The gradient is a blurred model . . . . .                             | 7         |
| 2.6      | Singular values, resolution and trade-off . . . . .                   | 7         |
| 2.7      | Local optimisation methods . . . . .                                  | 8         |
| <b>3</b> | <b>Wave physics and parameterizations</b>                             | <b>8</b>  |
| 3.1      | Variable-density acoustics . . . . .                                  | 8         |
| 3.2      | Isotropic elasticity . . . . .                                        | 8         |
| 3.3      | Anisotropy . . . . .                                                  | 9         |
| 3.4      | Attenuation . . . . .                                                 | 9         |
| 3.5      | Changes of variables . . . . .                                        | 10        |
| <b>4</b> | <b>The adjoint-state gradient</b>                                     | <b>10</b> |
| 4.1      | Derivation with a Lagrangian . . . . .                                | 10        |
| 4.2      | Acoustic kernels . . . . .                                            | 11        |
| 4.3      | Elastic kernels: strain and acceleration imaging conditions . . . . . | 11        |
| 4.4      | Time-domain implementation . . . . .                                  | 12        |
| 4.5      | Discretise-then-optimise . . . . .                                    | 12        |
| 4.6      | Source wavelet and data weighting . . . . .                           | 12        |
| <b>5</b> | <b>The Hessian</b>                                                    | <b>13</b> |
| 5.1      | Gauss–Newton and second-order terms . . . . .                         | 13        |
| 5.2      | Hessian-vector products by second-order adjoints . . . . .            | 13        |
| 5.3      | Finite-difference Hessian-vector products . . . . .                   | 14        |
| 5.4      | Structure of the multiparameter Hessian . . . . .                     | 14        |
| 5.5      | Pseudo-Hessians and illumination . . . . .                            | 15        |
| <b>6</b> | <b>Scattering theory and radiation patterns</b>                       | <b>16</b> |
| 6.1      | The Born approximation . . . . .                                      | 16        |
| 6.2      | Acoustic radiation patterns . . . . .                                 | 16        |
| 6.3      | Elastic radiation patterns . . . . .                                  | 17        |
| 6.4      | Reading the elastic patterns . . . . .                                | 17        |
| 6.5      | Radiation patterns are AVO coefficients . . . . .                     | 18        |
| 6.6      | Scattering angle and wavenumber . . . . .                             | 19        |
| 6.7      | Anisotropic radiation patterns . . . . .                              | 19        |
| <b>7</b> | <b>Cross-talk: definition, origin and quantification</b>              | <b>20</b> |
| 7.1      | The local Hessian kernel . . . . .                                    | 20        |
| 7.2      | What the numbers say . . . . .                                        | 20        |
| 7.3      | Sensitivity imbalance versus coupling . . . . .                       | 21        |
| 7.4      | Coupling, trade-off and posterior correlation . . . . .               | 22        |
| 7.5      | Leakage from modelling errors . . . . .                               | 22        |

|           |                                                                                  |           |
|-----------|----------------------------------------------------------------------------------|-----------|
| 7.6       | Measuring cross-talk in practice . . . . .                                       | 23        |
| <b>8</b>  | <b>Techniques for resolving cross-talk</b>                                       | <b>23</b> |
| 8.1       | Choosing the parameterization . . . . .                                          | 23        |
| 8.2       | Scaling and normalisation . . . . .                                              | 24        |
| 8.3       | Diagonal preconditioning and its limits . . . . .                                | 24        |
| 8.4       | Block-diagonal (point-wise $N_p \times N_p$ ) preconditioning . . . . .          | 24        |
| 8.5       | Quasi-Newton methods . . . . .                                                   | 25        |
| 8.6       | Truncated Newton (Newton-CG) . . . . .                                           | 25        |
| 8.7       | Hierarchical and multi-stage strategies . . . . .                                | 26        |
| 8.8       | Data selection and weighting . . . . .                                           | 27        |
| 8.9       | Multicomponent data and converted waves . . . . .                                | 27        |
| 8.10      | Regularisation, priors and constraints . . . . .                                 | 27        |
| 8.11      | Model-domain deblurring and subspace methods . . . . .                           | 28        |
| 8.12      | Summary of remedies . . . . .                                                    | 28        |
| <b>9</b>  | <b>A numerical laboratory</b>                                                    | <b>28</b> |
| 9.1       | Set-up: linearised elastic AVO as Born inversion . . . . .                       | 28        |
| 9.2       | Single-scatterer leakage . . . . .                                               | 29        |
| 9.3       | Updates and convergence with a band-limited wavelet . . . . .                    | 29        |
| 9.4       | The limits: aperture, noise and damping . . . . .                                | 30        |
| <b>10</b> | <b>Implementation guidance</b>                                                   | <b>31</b> |
| 10.1      | Verification tests . . . . .                                                     | 31        |
| 10.2      | Choosing the finite-difference step . . . . .                                    | 32        |
| 10.3      | Cost accounting . . . . .                                                        | 33        |
| 10.4      | A recommended workflow . . . . .                                                 | 33        |
| <b>11</b> | <b>Mapping the theory onto <code>tdfwi-hess</code></b>                           | <b>33</b> |
| 11.1      | What the current implementation does, in the language of this tutorial . . . . . | 34        |
| 11.2      | Natural next steps suggested by the theory . . . . .                             | 34        |
| <b>12</b> | <b>Summary</b>                                                                   | <b>35</b> |
| <b>A</b>  | <b>Notation</b>                                                                  | <b>36</b> |
| <b>B</b>  | <b>Derivation of the Born scattering amplitude</b>                               | <b>36</b> |
| <b>C</b>  | <b>The l-BFGS two-loop recursion</b>                                             | <b>36</b> |
| <b>D</b>  | <b>Exercises</b>                                                                 | <b>37</b> |

# 1 Introduction

## 1.1 Why multiparameter FWI?

Full-waveform inversion was formulated in the early 1980s as a least-squares fit of complete seismograms (Lailly, 1983; Tarantola, 1984). It has since become a production tool for building high-resolution P-wave velocity models (Virieux and Operto, 2009; Warner et al., 2013). In its most widespread form, FWI inverts for a *single* parameter class, usually the P-wave velocity  $V_P$  in an acoustic (possibly anisotropic) medium, while the other properties are fixed by empirical relations or by earlier processing. That simplification is powerful but limited:

- The Earth is elastic and anisotropic, and it attenuates. If the simulator ignores physics that is present in the data, the inversion compensates with spurious structure in the parameters it does invert for.
- Reservoir characterisation, CO<sub>2</sub> monitoring and geohazard assessment need quantities such as  $V_P/V_S$ , Poisson’s ratio, impedances and density. These are inaccessible to single-parameter inversion.
- Multicomponent (ocean-bottom node, land 3C) data record converted and shear waves whose kinematics and amplitudes depend directly on  $V_S$  and density.

*Multiparameter* FWI therefore inverts several parameter classes simultaneously, for example  $\mathbf{m} = (V_P, V_S, \rho)$  in isotropic elasticity,  $(V_P, \epsilon, \delta)$  in acoustic vertical transverse isotropy, or  $(V_P, Q_P)$  in visco-acoustic media. Good reviews are Virieux and Operto (2009), Operto et al. (2013), Yang et al. (2016) and Métivier et al. (2017).

## 1.2 The cross-talk problem in one paragraph

A small perturbation of any one parameter at a point  $\mathbf{x}$  scatters an incident wave in an angular pattern called the *radiation pattern* (Section 6). If two parameters have radiation patterns that are nearly proportional over the range of scattering angles the acquisition samples, their data signatures are nearly indistinguishable. The data then constrain only a combination of the two parameters, and an update aimed at one leaks into the other. Mathematically, the leakage is carried by the off-diagonal blocks of the multiparameter Hessian (Section 5). The gradient, which is the update direction of every first-order method, is the true model perturbation blurred by that Hessian. As a result, the gradient *always* contains cross-talk unless the Hessian is block-diagonal. Cross-talk is thus not a bug in a particular code. It is a property of the physics, the parameterization, the acquisition and the optimisation algorithm, and it must be addressed at all four levels.

## 1.3 Three faces of cross-talk

It helps to distinguish three mechanisms that are often lumped together (Section 7):

1. **Coupling (Hessian cross-talk).** The off-diagonal blocks of  $\mathbf{H}$  are non-zero, so the gradient for parameter  $p$  contains contributions from true perturbations of parameter  $q$ . A good approximation of the inverse Hessian removes this, *provided* the problem is well-conditioned.
2. **Trade-off (null-space cross-talk).** The radiation patterns are (nearly) linearly dependent over the available aperture, so a combination of parameters is (almost) invisible in the data. No optimiser can resolve it. Only extra information can: more aperture, other wave modes, regularisation, prior knowledge, or a parameterization that confines the ambiguity to one parameter.

3. **Leakage from modelling errors.** Physics missing from the simulator (attenuation, anisotropy, elasticity, source-wavelet errors) produces residuals that the inversion explains with the parameters it has.

## 1.4 What this tutorial contains

The tutorial is written for a reader comfortable with linear algebra, calculus and the basics of wave propagation who wants to understand, and implement, multiparameter FWI. It is organised as follows.

- Section 2 reviews least-squares inverse theory: Fréchet derivatives, adjoints, normal equations, resolution, regularisation and local optimisation.
- Section 3 covers wave physics and parameterizations: acoustic, elastic, anisotropic and attenuating media, and changes of variables.
- Section 4 derives the adjoint-state gradient, including the elastic strain–acceleration imaging conditions that production codes implement.
- Section 5 treats the Hessian: the Gauss–Newton and full Hessians, second-order adjoints, finite-difference Hessian-vector products, point-spread functions and the block structure.
- Section 6 develops Born scattering, radiation patterns, the link to amplitude-versus-angle (AVO) analysis, and wavenumber illumination.
- Section 7 defines and quantifies cross-talk.
- Section 8 is a catalogue of remedies with their strengths and weaknesses.
- Section 9 is a numerical laboratory that demonstrates those remedies quantitatively.
- Section 10 gives implementation guidance and Section 11 connects everything to the `tdfwi-hess` code.

### How to read this document

Boxes like this one summarise key results. Orange boxes give practical advice. Every figure and every number in the tables is produced by `make_figures.py` in this directory (a few seconds on a CPU); its printed log is saved in `figures/numbers.txt`. Readers in a hurry can read Sections 6–9 first and return to the derivations as needed.

## 1.5 Notation

Bold lower case denotes vectors (models  $\mathbf{m}$ , wavefields  $\mathbf{u}$ , data  $\mathbf{d}$ ). Bold upper case denotes matrices and linear operators.  $^\top$  is the transpose and  $^\dagger$  the adjoint with respect to the relevant inner product,  $\langle \cdot, \cdot \rangle$ . The forward modelling operator is  $\mathcal{F}(\mathbf{m})$ , the wave operator  $\mathcal{A}(\mathbf{m})$ , the receiver restriction  $\mathcal{R}$ , the Jacobian (Fréchet derivative, Born operator)  $\mathbf{L} = \partial \mathcal{F} / \partial \mathbf{m}$ , the gradient  $\mathbf{g}$ , the Hessian  $\mathbf{H}$  and the Gauss–Newton Hessian  $\mathbf{H}_{\text{GN}} = \mathbf{L}^\dagger \mathbf{L}$ . Parameters are indexed by  $p, q \in \{1, \dots, N_p\}$  and spatial positions by  $\mathbf{x}$ . Appendix A collects all symbols.

## 2 Foundations: least-squares inverse problems

### 2.1 The forward problem as an operator

Let  $\mathbf{m} \in \mathcal{M}$  be a model, a vector holding  $N_p$  parameter fields on a grid of  $N_x$  points, so that  $\dim \mathcal{M} = N_p N_x$ . We write it in block form

$$\mathbf{m} = \begin{pmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \vdots \\ \mathbf{m}_{N_p} \end{pmatrix}, \quad \mathbf{m}_p \in \mathbb{R}^{N_x}. \quad (1)$$

For each source  $s$ , the wavefield  $\mathbf{u}_s$  solves a linear wave equation whose coefficients depend on  $\mathbf{m}$ ,

$$\mathcal{A}(\mathbf{m}) \mathbf{u}_s = \mathbf{f}_s, \quad (2)$$

and the predicted data are the wavefield sampled at the receivers,  $\mathcal{F}_s(\mathbf{m}) = \mathcal{R}_s \mathbf{u}_s(\mathbf{m})$ . The forward map  $\mathcal{F} : \mathcal{M} \rightarrow \mathcal{D}$  is nonlinear in  $\mathbf{m}$  even though (2) is linear in  $\mathbf{u}$ , because  $\mathbf{u} = \mathcal{A}(\mathbf{m})^{-1} \mathbf{f}$ .

### 2.2 Misfit functions and their statistical meaning

The classical FWI objective is the least-squares misfit

$$J(\mathbf{m}) = \frac{1}{2} \sum_s \left\| \mathbf{W}_d (\mathcal{F}_s(\mathbf{m}) - \mathbf{d}_s^{\text{obs}}) \right\|^2 + \frac{\beta}{2} \left\| \mathbf{W}_m (\mathbf{m} - \mathbf{m}_{\text{prior}}) \right\|^2, \quad (3)$$

with data weighting  $\mathbf{W}_d$  and an optional model regularisation term. Under Gaussian assumptions, with data covariance  $\mathbf{C}_d = (\mathbf{W}_d^T \mathbf{W}_d)^{-1}$  and prior covariance  $\mathbf{C}_m = (\beta \mathbf{W}_m^T \mathbf{W}_m)^{-1}$ ,  $J$  is the negative log posterior (Tarantola, 2005). Near the minimum, the inverse of the Hessian of  $J$  is then the posterior covariance of the model,

$$\mathbf{C}_{\text{post}} \approx \mathbf{H}^{-1}. \quad (4)$$

This is the first hint that multiparameter cross-talk, which lives in the off-diagonal blocks of  $\mathbf{H}$ , is also a statement about *posterior correlation* between parameter classes. Other misfits (cross-correlation traveltimes, envelope, optimal transport, normalised or adaptive matching) change the data term, but they do not change the structure of what follows. We use the residual  $\mathbf{r}_s = \mathcal{F}_s(\mathbf{m}) - \mathbf{d}_s^{\text{obs}}$  and drop the source index when it is not needed.

### 2.3 Linearisation: Fréchet derivative and adjoint

The Fréchet derivative (Jacobian, sensitivity, Born operator) of  $\mathcal{F}$  at  $\mathbf{m}$  is the linear operator  $\mathbf{L}(\mathbf{m})$  with

$$\mathcal{F}(\mathbf{m} + \delta \mathbf{m}) = \mathcal{F}(\mathbf{m}) + \mathbf{L}(\mathbf{m}) \delta \mathbf{m} + \mathcal{O}(\|\delta \mathbf{m}\|^2). \quad (5)$$

In block form,  $\mathbf{L} = [\mathbf{L}_1 \ \mathbf{L}_2 \ \cdots \ \mathbf{L}_{N_p}]$ , where  $\mathbf{L}_p$  maps a perturbation of parameter class  $p$  to data. The adjoint  $\mathbf{L}^\dagger$  is defined by  $\langle \mathbf{L} \delta \mathbf{m}, \mathbf{d} \rangle_{\mathcal{D}} = \langle \delta \mathbf{m}, \mathbf{L}^\dagger \mathbf{d} \rangle_{\mathcal{M}}$  for all  $\delta \mathbf{m}, \mathbf{d}$ . With the Euclidean inner product,  $\mathbf{L}^\dagger = \mathbf{L}^T$ . Physically,  $\mathbf{L}$  is *Born modelling*: secondary sources excited by  $\delta \mathbf{m}$  are propagated to the receivers.  $\mathbf{L}^\dagger$  is *migration*: data are back-propagated and correlated with the incident field. The gradient of the data term of (3) (with  $\mathbf{W}_d = \mathbf{I}$ ) is

$$\mathbf{g} = \nabla_{\mathbf{m}} J = \mathbf{L}^\dagger \mathbf{r}, \quad \mathbf{g}_p = \mathbf{L}_p^\dagger \mathbf{r}. \quad (6)$$

The gradient for parameter  $p$  is therefore the residual migrated with parameter  $p$ 's imaging condition.

## 2.4 Second derivatives and the Newton equation

Differentiating (6) once more gives the Hessian

$$\mathbf{H} = \nabla_{\mathbf{m}}^2 J = \underbrace{\mathbf{L}^\dagger \mathbf{L}}_{\mathbf{H}_{\text{GN}}} + \underbrace{\sum_i r_i \nabla_{\mathbf{m}}^2 \mathcal{F}_i}_{\mathbf{Q}}, \quad (7)$$

where  $i$  runs over all data samples.  $\mathbf{H}_{\text{GN}}$  is the Gauss–Newton (GN) Hessian. It is symmetric positive semi-definite and involves only first derivatives.  $\mathbf{Q}$  is the second-order term: it is weighted by the residual, contains double-scattering information and can make  $\mathbf{H}$  indefinite. A local quadratic model of  $J$  around  $\mathbf{m}_k$  is

$$J(\mathbf{m}_k + \delta \mathbf{m}) \approx J(\mathbf{m}_k) + \mathbf{g}_k^\top \delta \mathbf{m} + \frac{1}{2} \delta \mathbf{m}^\top \mathbf{H}_k \delta \mathbf{m}, \quad (8)$$

whose minimiser solves the *Newton equation*

$$\mathbf{H}_k \delta \mathbf{m} = -\mathbf{g}_k. \quad (9)$$

Replacing  $\mathbf{H}$  by  $\mathbf{H}_{\text{GN}}$  gives the Gauss–Newton step, and adding damping  $(\mathbf{H}_{\text{GN}} + \mu \mathbf{I}) \delta \mathbf{m} = -\mathbf{g}$  gives the Levenberg–Marquardt step. For a linear problem ( $\mathcal{F}(\mathbf{m}) = \mathbf{Lm}$ ), the Gauss–Newton step from  $\mathbf{m} = 0$  is the solution of the *normal equations*  $\mathbf{L}^\top \mathbf{L} \mathbf{m} = \mathbf{L}^\top \mathbf{d}$ .

## 2.5 The gradient is a blurred model

Suppose the data are exactly linear,  $\mathbf{d}^{\text{obs}} = \mathbf{L} \delta \mathbf{m}_{\text{true}}$ , and we start from the background ( $\delta \mathbf{m} = 0$ ). Then  $\mathbf{r} = -\mathbf{L} \delta \mathbf{m}_{\text{true}}$  and

$$-\mathbf{g} = \mathbf{L}^\dagger \mathbf{L} \delta \mathbf{m}_{\text{true}} = \mathbf{H}_{\text{GN}} \delta \mathbf{m}_{\text{true}}, \quad -\mathbf{g}_p = \sum_q \mathbf{H}_{\text{GN}}^{(pq)} \delta \mathbf{m}_{q,\text{true}}. \quad (10)$$

This short identity is the root of everything in this tutorial. The descent direction for parameter  $p$  is the true perturbation of  $p$  blurred by the diagonal block  $\mathbf{H}_{\text{GN}}^{(pp)}$ , *plus* the true perturbations of all other parameters blurred by the off-diagonal blocks  $\mathbf{H}_{\text{GN}}^{(pq)}$ . The off-diagonal contributions are the cross-talk. Steepest descent, nonlinear conjugate gradients and any method that uses the raw gradient inherit it at the first iteration. Newton’s method removes it because  $\delta \mathbf{m} = -\mathbf{H}^{-1} \mathbf{g} = \delta \mathbf{m}_{\text{true}}$ , provided  $\mathbf{H}$  is invertible.

## 2.6 Singular values, resolution and trade-off

Let  $\mathbf{L} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$  be the singular value decomposition. Model-space singular vectors  $\mathbf{v}_j$  with large singular values  $\sigma_j$  are well resolved. Those with small  $\sigma_j$  are poorly resolved, and those with  $\sigma_j = 0$  span the null space. The damped least-squares solution

$$\delta \mathbf{m}_\mu = (\mathbf{H}_{\text{GN}} + \mu \mathbf{I})^{-1} \mathbf{L}^\top \mathbf{d} = \sum_j \frac{\sigma_j}{\sigma_j^2 + \mu} (\mathbf{u}_j^\top \mathbf{d}) \mathbf{v}_j \quad (11)$$

filters out directions with  $\sigma_j^2 \ll \mu$ . The *model resolution matrix*

$$\mathbf{R}_\mu = (\mathbf{H}_{\text{GN}} + \mu \mathbf{I})^{-1} \mathbf{H}_{\text{GN}}, \quad \delta \mathbf{m}_\mu = \mathbf{R}_\mu \delta \mathbf{m}_{\text{true}} \text{ (noise-free)}, \quad (12)$$

generalises (10):  $\mathbf{R}_\mu \rightarrow \mathbf{I}$  when  $\mu \rightarrow 0$  and  $\mathbf{H}_{\text{GN}}$  is invertible, and  $\mathbf{R}_\mu \approx \mu^{-1} \mathbf{H}_{\text{GN}}$  (the gradient) when  $\mu$  is large. The columns of  $\mathbf{R}_\mu$  are point-spread functions. Their off-diagonal blocks quantify the cross-talk that *remains* after inversion. In multiparameter problems, a singular vector  $\mathbf{v}_j$  with small  $\sigma_j$  is typically a *mixture* of parameter classes, for instance  $\mathbf{v} \propto (\delta V_P/V_P, \delta \rho/\rho) \approx (1, -1)/\sqrt{2}$ , which leaves  $I_P = \rho V_P$  unchanged. Such a direction cannot be determined from the data. This is *trade-off* (Section 7).

## 2.7 Local optimisation methods

FWI problems have  $10^6$ – $10^{10}$  unknowns, so  $\mathbf{H}$  can never be formed explicitly. Practical methods iterate  $\mathbf{m}_{k+1} = \mathbf{m}_k + \alpha_k \mathbf{s}_k$ , with a search direction  $\mathbf{s}_k$  and a step length  $\alpha_k$  that satisfies the Wolfe conditions (Nocedal and Wright, 2006).

**Steepest descent (SD)**  $\mathbf{s}_k = -\mathbf{P}\mathbf{g}_k$ , with a preconditioner  $\mathbf{P}$ . Cheap, but convergence is governed by  $\text{cond}(\mathbf{P}\mathbf{H})$  and the update carries the full cross-talk of (10).

**Nonlinear conjugate gradients (NLCG)**  $\mathbf{s}_k = -\mathbf{P}\mathbf{g}_k + \beta_k \mathbf{s}_{k-1}$ , with the Fletcher–Reeves or Polak–Ribière  $\beta_k$ . Implicitly builds Krylov information about  $\mathbf{H}$  across iterations.

**Quasi-Newton (l-BFGS)**  $\mathbf{s}_k = -\mathbf{B}_k^{-1} \mathbf{g}_k$ , where  $\mathbf{B}_k^{-1}$  is a low-rank-updated inverse-Hessian approximation built from the last  $\ell$  pairs  $(\mathbf{m}_{j+1} - \mathbf{m}_j, \mathbf{g}_{j+1} - \mathbf{g}_j)$  (Liu and Nocedal, 1989; Nocedal, 1980). It captures some off-diagonal curvature and therefore partially corrects cross-talk.

**Truncated Newton (TN, Newton-CG)** approximately solves (9) by an inner conjugate gradient loop that needs only Hessian-vector products  $\mathbf{H}\mathbf{v}$  (Nash, 2000; Métivier et al., 2013). This is the most direct way of applying  $\mathbf{H}^{-1}$ , and therefore of removing coupling cross-talk (Section 8.6).

### Key result

The gradient is the true model blurred by the Hessian,  $-\mathbf{g} = \mathbf{H}_{\text{GN}} \delta \mathbf{m}_{\text{true}}$ . Its off-diagonal blocks  $\mathbf{H}_{\text{GN}}^{(pq)} = \mathbf{L}_p^\dagger \mathbf{L}_q$  inject parameter  $q$  into the update of parameter  $p$ . Any method that approximates  $\mathbf{H}^{-1}$  well reduces this coupling. No method can resolve directions in the (near) null space of  $\mathbf{L}$ .

## 3 Wave physics and parameterizations

The parameter classes we can invert for are fixed by the wave equation we simulate. The ones we *should* invert for depend on which combinations the data resolve. This section sets out the equations and the parameterizations used in practice.

### 3.1 Variable-density acoustics

For pressure  $p(\mathbf{x}, t)$  in a fluid with bulk modulus  $\kappa = \rho V_P^2$  and density  $\rho$ ,

$$\frac{1}{\kappa} \partial_t^2 p - \nabla \cdot \left( \frac{1}{\rho} \nabla p \right) = s. \quad (13)$$

Two parameters enter: the *compressibility*  $1/\kappa$  multiplies the time derivative, and the *buoyancy*  $b = 1/\rho$  sits inside the spatial divergence. These two places in the equation produce the two distinct radiation patterns of Section 6: a monopole for  $\kappa$  and a dipole for  $\rho$ . Common parameterizations are  $(\kappa, \rho)$ ,  $(V_P, \rho)$ ,  $(I_P = \rho V_P, \rho)$  and  $(V_P, I_P)$ . With constant density, (13) reduces to the constant-density wave equation that most production FWI uses. Its single parameter is  $V_P$ , and its neglect of density is itself a source of leakage (Section 7.5).

### 3.2 Isotropic elasticity

The elastodynamic equations for displacement  $\mathbf{u}(\mathbf{x}, t)$  read

$$\rho \partial_t^2 u_i - \partial_j (c_{ijkl} \partial_l u_k) = f_i, \quad c_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (14)$$

Table 1: Common parameterizations for isotropic elastic FWI. The physical arguments are summarised in Sections 6–8.

| Class                    | Parameters                                         | Remarks                                                                                                                                                                            |
|--------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Velocity                 | $(V_P, V_S, \rho)$                                 | Kinematically natural. $V_P$ is well resolved by transmitted and diving waves. Strong $V_P$ – $\rho$ trade-off in short-offset reflections.                                        |
| Impedance                | $(I_P, I_S, \rho)$                                 | $I_P = \rho V_P$ , $I_S = \rho V_S$ . Near-normal reflections depend on $I_P$ only, which decouples density from $I_P$ there. Density is then poorly resolved but no longer leaks. |
| Lam $\tilde{\text{A}}$ l | $(\lambda, \mu, \rho)$                             | Appear directly in (14). Their patterns are orthogonal to density’s over the full aperture, but strongly coupled for limited apertures.                                            |
| Mixed                    | $(V_P, I_P, V_S), (V_P, V_P/V_S, I_P^{-1}), \dots$ | Designed to isolate the poorly constrained parameter or the rock-physics quantity of interest.                                                                                     |
| Logarithmic              | $(\ln V_P, \ln V_S, \ln \rho)$                     | Dimensionless. Gradients are relative-perturbation sensitivities, which provides automatic scaling (Section 8.2).                                                                  |

with Lam $\tilde{\text{A}}$ l parameters  $\lambda = \rho(V_P^2 - 2V_S^2)$  and  $\mu = \rho V_S^2$ . Equivalently, in first-order velocity–stress form (Virieux, 1986; Levander, 1988), which is the form most staggered-grid codes implement,

$$\rho \partial_t \mathbf{v} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}, \quad \partial_t \boldsymbol{\sigma} = \lambda (\nabla \cdot \mathbf{v}) \mathbf{I} + \mu (\nabla \mathbf{v} + \nabla \mathbf{v}^T). \quad (15)$$

Three independent parameters appear. Common choices are summarised in Table 1. The strain tensor is  $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ , and  $\nabla \cdot \mathbf{u} = \text{tr } \boldsymbol{\varepsilon}$  is the dilatation. These two quantities will appear in the imaging conditions of Section 4.

### 3.3 Anisotropy

Sedimentary rocks are commonly transversely isotropic. For vertical symmetry axis (VTI), the stiffness in Voigt notation is characterised by  $C_{33} = \rho V_{P0}^2$ ,  $C_{44} = \rho V_{S0}^2$  and Thomsen’s dimensionless parameters (Thomsen, 1986)

$$\epsilon = \frac{C_{11} - C_{33}}{2C_{33}}, \quad \delta = \frac{(C_{13} + C_{44})^2 - (C_{33} - C_{44})^2}{2C_{33}(C_{33} - C_{44})}, \quad \gamma_T = \frac{C_{66} - C_{44}}{2C_{44}}. \quad (16)$$

Inverting the definition of  $\delta$  gives  $C_{13} = \sqrt{(C_{33} - C_{44})(C_{33}(1 + 2\delta) - C_{44})} - C_{44}$ , which is the form used in VTI/TTI imaging conditions. For tilted symmetry axes (TTI) the stiffness is rotated by the tilt and azimuth angles, which are usually fixed from structural dips rather than inverted. Pseudo-acoustic approximations, obtained by setting the shear velocity on the symmetry axis to zero (Alkhalifah, 1998), give acoustic anisotropic FWI in  $(V_{P0}, \epsilon, \delta)$  or equivalents such as the NMO velocity  $V_n = V_{P0}\sqrt{1 + 2\delta}$ , the horizontal velocity  $V_h = V_{P0}\sqrt{1 + 2\epsilon}$  and  $\eta = (\epsilon - \delta)/(1 + 2\delta)$ . Parameterization studies (Plessix and Cao, 2011; Gholami et al., 2013; Alkhalifah and Plessix, 2014; Kazei and Alkhalifah, 2019) show that anisotropic parameters have radiation patterns concentrated at particular scattering angles. For example,  $\delta$  affects mainly intermediate-angle reflection amplitudes and very little the kinematics of near-vertical waves. As a result, anisotropic FWI exhibits the same cross-talk structure as the elastic case, with different patterns.

### 3.4 Attenuation

Visco-acoustic or visco-elastic media are modelled by complex frequency-dependent moduli (for example a constant- $Q$  model) or, in the time domain, by a superposition of standard linear

solids with memory variables. The quality factor  $Q$  alters amplitudes and introduces dispersion (a frequency-dependent phase velocity). In the frequency domain,  $1/Q$  enters as an imaginary part of the modulus, a perturbation of which scatters with a pattern similar to that of  $V_P$  but phase-shifted by  $90^\circ$ . Velocity–attenuation cross-talk is therefore weak at a single frequency but becomes significant over a band because of dispersion (Kamei and Pratt, 2013; Keating and Innanen, 2019).

### 3.5 Changes of variables

Let  $\mathbf{m} = \phi(\tilde{\mathbf{m}})$  be a smooth, pointwise invertible map from new parameters  $\tilde{\mathbf{m}}$  to old ones (for example  $(I_P, I_S, \rho) \mapsto (V_P, V_S, \rho)$ ), with point-wise Jacobian  $\mathbf{T} = \partial\mathbf{m}/\partial\tilde{\mathbf{m}}$  (block-diagonal in space,  $N_p \times N_p$  at every point). The chain rule gives

$$\tilde{\mathbf{g}} = \mathbf{T}^\top \mathbf{g}, \quad \tilde{\mathbf{L}} = \mathbf{L}\mathbf{T}, \quad \tilde{\mathbf{H}}_{\text{GN}} = \mathbf{T}^\top \mathbf{H}_{\text{GN}} \mathbf{T}, \quad \tilde{\mathbf{H}} = \mathbf{T}^\top \mathbf{H} \mathbf{T} + \sum_p g_p \nabla_{\tilde{\mathbf{m}}}^2 \phi_p. \quad (17)$$

The Gauss–Newton Hessian transforms by *congruence*. By Sylvester’s law of inertia, congruence preserves the number of positive, zero and negative eigenvalues, so the null space cannot be removed by reparameterization: its dimension is invariant. Congruence does, however, change the *eigenvalues*, the *eigenvectors* and hence the *off-diagonal structure*. A good parameterization makes  $\tilde{\mathbf{H}}$  closer to block-diagonal and better conditioned, so that a gradient-based method suffers less cross-talk. A poorly constrained combination can also be aligned with a single parameter axis, so that the ambiguity is confined to one parameter instead of contaminating all of them (Section 8.1). For velocity and impedance in relative units,

$$\begin{pmatrix} \delta V_P/V_P \\ \delta V_S/V_S \\ \delta \rho/\rho \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_{\mathbf{T}} \begin{pmatrix} \delta I_P/I_P \\ \delta I_S/I_S \\ \delta \rho/\rho \end{pmatrix}. \quad (18)$$

#### Practical note: model vs inversion parameters

Production codes typically store and propagate in one fixed set of *model* parameters (for example  $V_P, V_S, \rho$ , written to disk and read by the propagator). They invert in a user-selected set of *inversion* parameters. The gradient is computed in the model parameters by the imaging condition and mapped by  $\mathbf{T}^\top$ . The update is mapped back by  $\phi$ . Any Hessian-vector product must use the same mapping. In `tdfwi-hess`, perturbed models for the finite-difference  $\mathbf{H}\mathbf{v}$  are built by the same `update_x` routine that applies inversion updates, which makes the Hessian-vector product exact for any of the supported parameterizations (Section 11).

## 4 The adjoint-state gradient

### 4.1 Derivation with a Lagrangian

Treat the wave equation (2) as a constraint and introduce a Lagrange multiplier (adjoint wavefield)  $\boldsymbol{\lambda}$ :

$$\mathcal{L}(\mathbf{u}, \boldsymbol{\lambda}, \mathbf{m}) = \frac{1}{2} \left\| \mathcal{R}\mathbf{u} - \mathbf{d}^{\text{obs}} \right\|^2 - \langle \boldsymbol{\lambda}, \mathcal{A}(\mathbf{m})\mathbf{u} - \mathbf{f} \rangle. \quad (19)$$

For  $\mathbf{u} = \mathbf{u}(\mathbf{m})$  satisfying the constraint,  $\mathcal{L} = J$  for every  $\boldsymbol{\lambda}$ , so  $dJ/d\mathbf{m} = \partial_{\mathbf{m}}\mathcal{L} + \partial_{\mathbf{u}}\mathcal{L} \partial_{\mathbf{m}}\mathbf{u}$ . Choosing  $\boldsymbol{\lambda}$  so that  $\partial_{\mathbf{u}}\mathcal{L} = 0$  eliminates the expensive term  $\partial_{\mathbf{m}}\mathbf{u}$  (Plessix, 2006):

$$\mathcal{A}(\mathbf{m})\mathbf{u} = \mathbf{f} \quad (\text{state / forward equation}) \quad (20)$$

$$\mathcal{A}(\mathbf{m})^\dagger \boldsymbol{\lambda} = \mathcal{R}^\top (\mathcal{R}\mathbf{u} - \mathbf{d}^{\text{obs}}) \quad (\text{adjoint equation}) \quad (21)$$

$$\nabla_{m_p(\mathbf{x})} J = - \left\langle \boldsymbol{\lambda}, \frac{\partial \mathcal{A}}{\partial m_p(\mathbf{x})} \mathbf{u} \right\rangle \quad (\text{gradient}) \quad (22)$$

For the time-domain wave equations above (self-adjoint in space, second order in time),  $\mathcal{A}^\dagger$  is the same wave operator run *backwards in time* from final conditions. The adjoint field is the data residual injected at the receivers and back-propagated. The gradient (22) is a zero-lag correlation of the back-propagated residual with the *virtual source*  $(\partial \mathcal{A} / \partial m_p) \mathbf{u}$  that a perturbation of parameter  $p$  would create. One gradient costs one forward and one adjoint simulation per source, *independently of the number of parameter classes*. All parameter gradients come from the same two wavefields, combined with different imaging conditions.

## 4.2 Acoustic kernels

For (13),  $\mathcal{A} = \kappa^{-1} \partial_t^2 - \nabla \cdot (b \nabla \cdot)$  with  $b = 1/\rho$ . Integrating by parts in time and space (with quiescent initial/final conditions and absorbing boundaries) gives

$$\nabla_\kappa J = -\frac{1}{\kappa^2} \int_0^T \partial_t \lambda \partial_t p \, dt, \quad \nabla_b J = -\int_0^T \nabla \lambda \cdot \nabla p \, dt. \quad (23)$$

(Sign conventions differ between references according to the sign of the adjoint source. Only consistency matters.) The chain rule with  $\kappa = \rho V_P^2$ ,  $b = 1/\rho$  gives the velocity–density kernels

$$\nabla_{V_P} J = 2\rho V_P \nabla_\kappa J, \quad \nabla_\rho J|_{V_P} = V_P^2 \nabla_\kappa J - \rho^{-2} \nabla_b J. \quad (24)$$

The two imaging conditions differ in kind. The  $\kappa$ -kernel correlates *time derivatives*, a scalar monopole correlation insensitive to the propagation directions of  $p$  and  $\lambda$ . The  $b$ -kernel correlates *spatial gradients*, so it is weighted by  $\cos \theta$  of the angle between the incident and back-propagated wave vectors. This angular weighting is the radiation pattern, seen from the imaging side.

## 4.3 Elastic kernels: strain and acceleration imaging conditions

For (14),  $\partial \mathcal{A} / \partial \rho = \partial_t^2$  and  $\partial \mathcal{A} / \partial c_{ijkl}$  acts through the strain. Integration by parts gives the kernels (Tarantola, 1986; Tromp et al., 2005; Fichtner, 2010)

$$\nabla_\lambda J = -\int (\nabla \cdot \boldsymbol{\lambda})(\nabla \cdot \mathbf{u}) \, dt, \quad \nabla_\mu J = -\int 2 \boldsymbol{\varepsilon}(\boldsymbol{\lambda}) : \boldsymbol{\varepsilon}(\mathbf{u}) \, dt, \quad \nabla_\rho J|_{\lambda, \mu} = -\int \boldsymbol{\lambda} \cdot \partial_t^2 \mathbf{u} \, dt. \quad (25)$$

Write  $g_\lambda = \langle \nabla \cdot \mathbf{u}_s, \nabla \cdot \mathbf{u}_r \rangle$  (dilatation  $\times$  dilatation),  $g_\mu = 2 \langle \boldsymbol{\varepsilon}(\mathbf{u}_s), \boldsymbol{\varepsilon}(\mathbf{u}_r) \rangle$  (full strain contraction) and  $g_\rho = \langle \mathbf{u}_s, \ddot{\mathbf{u}}_r \rangle$  (acceleration correlation), where  $\mathbf{u}_s$  is the source-side field and  $\mathbf{u}_r = \boldsymbol{\lambda}$  the receiver-side (adjoint) field, and the overall sign is absorbed. The chain rule then gives the velocity–density gradients

$$\begin{aligned} g_{V_P} &= 2\rho V_P g_\lambda, \\ g_{V_S} &= 2\rho V_S (g_\mu - 2g_\lambda), \\ g_\rho|_{V_P, V_S} &= g_\rho + (V_P^2 - 2V_S^2) g_\lambda + V_S^2 g_\mu. \end{aligned} \quad (26)$$

The companion SymPy study (docs/SymPy\_EFWI) proves that the isotropic limit of the production CUDA kernel in `tdfwi/image_condition_mp_efwi.cu` is symbolically identical to (26). The VTI kernel follows the same pattern, with the stiffness built from  $(C_{33}, C_{44}, \epsilon, \delta)$  through (16).

Several structural facts about (26) explain much of elastic cross-talk:

- $g_{V_P}$  depends only on the dilatation, i.e. on P-waves at both ends. It is blind to converted and shear energy.
- $g_{V_S}$  contains  $-2g_\lambda$ , i.e. *the  $V_P$  imaging condition itself*. Even a pure P-to-P residual produces a  $V_S$  gradient whenever  $g_\mu \neq 2g_\lambda$ . This is coupling at the level of the imaging condition, before any acquisition effect.
- $g_\rho$  contains both  $g_\lambda$  and  $g_\mu$  terms, so the density gradient mixes the kinetic (acceleration) and elastic (strain) contributions. Their relative weight depends on the scattering angle: they nearly cancel in forward scattering (Section 6).

Other parameterizations follow from (17). For impedances, with  $V_P = I_P/\rho$  and  $V_S = I_S/\rho$ ,

$$g_{I_P} = \frac{g_{V_P}}{\rho}, \quad g_{I_S} = \frac{g_{V_S}}{\rho}, \quad g_\rho|_{I_P, I_S} = g_\rho|_{V_P, V_S} - \frac{V_P}{\rho} g_{V_P} - \frac{V_S}{\rho} g_{V_S}. \quad (27)$$

#### 4.4 Time-domain implementation

Equation (22) correlates a forward field running from  $t = 0$  to  $T$  with an adjoint field running from  $T$  to 0. This requires one of the following:

1. **Storage.** Save the forward wavefield (or the needed strains and accelerations) every  $j_t$  time steps on disk or in host memory. With  $j_t$  near the Nyquist limit of the wavefield, this is the simplest and most common approach on GPUs.
2. **Checkpointing.** Store a few full states and recompute the intervals between them during the backward pass. The optimal **revolve** schedule trades memory against a logarithmic number of recomputations (Griewank and Walther, 2000; Symes, 2007).
3. **Reconstruction.** Store boundary values and run the forward field backwards in time alongside the adjoint field. This is exact for non-dissipative interior equations, but unstable with attenuation.

For multiparameter elastic FWI the stored quantities are the full strain tensor (six components in 3-D) and the particle acceleration or velocity. This is more memory than the single dilatation needed for  $V_P$  alone, which is a practical cost of the extra parameter classes.

#### 4.5 Discretise-then-optimize

The continuous adjoint equations (20)–(22) are exact for the *continuous* problem. A code that discretises them independently (“optimize-then-discretise”) computes a gradient that matches the discrete objective only to discretisation accuracy. Mismatches are harmless for steepest descent but matter for quasi-Newton methods and line searches, and above all for Hessian-vector products formed by finite differences of gradients (Section 5.3). There, an inconsistent gradient produces an asymmetric “Hessian”. The remedy is to verify the discrete gradient against finite differences of the discrete objective (Section 10.1). Where it matters, use the exact discrete adjoint of the propagator (“discretise-then-optimize”), including boundary conditions, source injection and receiver interpolation.

#### 4.6 Source wavelet and data weighting

The source signature is usually unknown and is estimated by a frequency-domain Wiener filter at each iteration (Pratt, 1999). Because amplitude information drives the separation of parameters (Section 6), multiparameter FWI is much more sensitive to wavelet and amplitude errors than

kinematic, velocity-only FWI. An incorrect wavelet amplitude spectrum leaks into impedance-type parameters and density. Data weighting  $\mathbf{W}_d$  (offset, time, component, or mute weights) changes the effective illumination and therefore the Hessian. It is one of the levers for controlling cross-talk (Section 8.8).

## 5 The Hessian

### 5.1 Gauss–Newton and second-order terms

The decomposition (7),  $\mathbf{H} = \mathbf{H}_{\text{GN}} + \mathbf{Q}$ , has a clean physical reading (Pratt et al., 1998; Fichtner and Trampert, 2011a).

- $\mathbf{H}_{\text{GN}}^{(pq)}(\mathbf{x}, \mathbf{x}') = \sum_{s,r} \int \overline{L_p(\mathbf{x}; s, r, t)} L_q(\mathbf{x}'; s, r, t) dt$  measures how much the single-scattered data generated by a perturbation of parameter  $q$  at  $\mathbf{x}'$  resemble those generated by parameter  $p$  at  $\mathbf{x}$ . Its diagonal ( $p = q$ ,  $\mathbf{x} = \mathbf{x}'$ ) is the *illumination* of parameter  $p$ . Its decay away from  $\mathbf{x} = \mathbf{x}'$  is the *spatial* resolution. Its off-diagonal blocks  $p \neq q$  are the *inter-parameter* coupling.
- $\mathbf{Q} = \sum_i r_i \nabla^2 \mathcal{F}_i$  couples model perturbations through second-order (double) scattering, weighted by the current residual. It vanishes for zero-residual problems. It is significant when the starting model is far from the truth or when the data contain strong multiple scattering (free-surface multiples, interbed multiples, salt bodies). It can be indefinite, as Figure 1c shows: the full Hessian of a simple layered problem acquires negative eigenvalues as soon as the starting-model error reaches about 30% of the reference perturbation.

### 5.2 Hessian-vector products by second-order adjoints

Newton-type methods need only the action  $\mathbf{H}\mathbf{v}$  for a given direction  $\mathbf{v}$ . Differentiating the gradient (22) in the direction  $\mathbf{v}$  gives the *second-order adjoint* recipe (Fichtner and Trampert, 2011a; Métivier et al., 2013). Write  $\mathcal{A}_p \equiv \partial \mathcal{A} / \partial m_p$  and  $\mathcal{A}_{\mathbf{v}} \equiv \sum_q v_q \mathcal{A}_q$ . Then:

1. **Background fields.** Solve  $\mathcal{A}\mathbf{u} = \mathbf{f}$  and  $\mathcal{A}^\dagger \boldsymbol{\lambda} = \mathcal{R}^\top \mathbf{r}$  (these are already available from the gradient).
2. **First-order (Born) field.** Solve  $\mathcal{A} \delta \mathbf{u} = -\mathcal{A}_{\mathbf{v}} \mathbf{u}$ . Then  $\mathcal{R} \delta \mathbf{u} = \mathbf{L}\mathbf{v}$  are the Born data of the perturbation  $\mathbf{v}$ .
3. **Second-order adjoint field.** Solve  $\mathcal{A}^\dagger \delta \boldsymbol{\lambda} = \mathcal{R}^\top \mathcal{R} \delta \mathbf{u} - \mathcal{A}_{\mathbf{v}}^\dagger \boldsymbol{\lambda}$ .
4. **Imaging.**

$$(\mathbf{H}\mathbf{v})_p = -\langle \delta \boldsymbol{\lambda}, \mathcal{A}_p \mathbf{u} \rangle - \langle \boldsymbol{\lambda}, \mathcal{A}_p \delta \mathbf{u} \rangle - \langle \boldsymbol{\lambda}, (\partial_{\mathbf{v}} \mathcal{A}_p) \mathbf{u} \rangle. \quad (28)$$

Dropping every term that contains the residual-driven field  $\boldsymbol{\lambda}$  (the source term  $-\mathcal{A}_{\mathbf{v}}^\dagger \boldsymbol{\lambda}$  in step 3 and the last two terms in (28)) leaves the Gauss–Newton product

$$(\mathbf{H}_{\text{GN}}\mathbf{v})_p = -\langle \delta \boldsymbol{\lambda}_{\text{GN}}, \mathcal{A}_p \mathbf{u} \rangle, \quad \mathcal{A}^\dagger \delta \boldsymbol{\lambda}_{\text{GN}} = \mathcal{R}^\top \mathbf{L}\mathbf{v}, \quad (29)$$

which is literally  $\mathbf{L}^\dagger(\mathbf{L}\mathbf{v})$ : Born modelling followed by migration with the same imaging conditions as the gradient. Either product costs two additional wave simulations per source (one forward and one adjoint) on top of the stored background fields. The cost is independent of  $N_p$ .

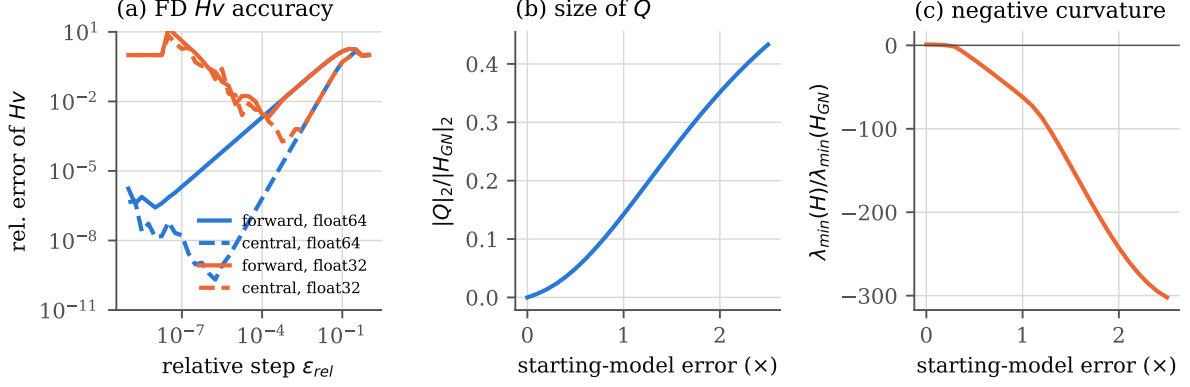


Figure 1: (a) Relative error of finite-difference Hessian-vector products versus the relative step  $\epsilon_{\text{rel}}$  (step scaled per parameter, Section 10.2) for a nonlinear 1-D layered-medium problem with 14 free velocity/density parameters. Gradients are computed by complex-step differentiation in float64 or float32, and the reference is a fourth-order difference in float64. Each curve shows the classic V shape: truncation error on the right, round-off on the left. (b) Relative size of the second-order term  $\mathbf{Q}$  as the starting model is degraded. (c) The smallest eigenvalue of the full Hessian, relative to that of  $\mathbf{H}_{\text{GN}}$ . Once the residual is non-negligible,  $\mathbf{H}$  becomes indefinite, with negative curvature up to  $300\times$  larger in magnitude than the smallest Gauss–Newton eigenvalue.

### 5.3 Finite-difference Hessian-vector products

When a code already computes accurate gradients but has no linearised (Born) propagator, the simplest Hessian-vector product is a directional finite difference of the gradient:

$$\mathbf{H}\mathbf{v} \approx \frac{\mathbf{g}(\mathbf{m} + \epsilon\mathbf{v}) - \mathbf{g}(\mathbf{m})}{\epsilon} \quad (\text{forward}), \quad \mathbf{H}\mathbf{v} \approx \frac{\mathbf{g}(\mathbf{m} + \epsilon\mathbf{v}) - \mathbf{g}(\mathbf{m} - \epsilon\mathbf{v})}{2\epsilon} \quad (\text{central}). \quad (30)$$

This gives the action of the *full* Hessian (including  $\mathbf{Q}$ ). The forward difference costs one extra gradient, i.e. two simulations, the same as (28). The central difference costs two extra gradients. The step  $\epsilon$  balances truncation error,  $\mathcal{O}(\epsilon)$  forward or  $\mathcal{O}(\epsilon^2)$  central, against round-off error,  $\mathcal{O}(\eta \|\mathbf{g}\| / \epsilon)$ , where  $\eta$  is the relative precision of the gradient. The optimal steps are

$$\epsilon_{\text{fwd}}^* \sim \sqrt{\eta}, \quad \epsilon_{\text{cen}}^* \sim \eta^{1/3}, \quad \text{error}_{\text{fwd}}^* \sim \sqrt{\eta}, \quad \text{error}_{\text{cen}}^* \sim \eta^{2/3}, \quad (31)$$

in units of a relative perturbation of the model. GPU propagators run in single precision ( $\eta \approx 6 \times 10^{-8}$  at best, effectively  $\eta \sim 10^{-6}$  after accumulation over many time steps). The best achievable accuracy is therefore  $\sim 10^{-3}$  for forward differences and  $\sim 10^{-4}$  for central differences. Figure 1a confirms this on a nonlinear layered-medium problem: in float32 the forward difference is best at  $\epsilon_{\text{rel}} \approx 2 \times 10^{-4}$  (error  $2 \times 10^{-3}$ ) and the central difference at  $\epsilon_{\text{rel}} \approx 6 \times 10^{-4}$  (error  $2 \times 10^{-4}$ ). Truncated-Newton methods tolerate such inexact products (Section 8.6). Section 10.2 explains why the step must be scaled *per parameter class*.

### 5.4 Structure of the multiparameter Hessian

Ordering the unknowns parameter-major, as in (1), the GN Hessian is an  $N_p \times N_p$  array of  $N_x \times N_x$  blocks:

$$\mathbf{H}_{\text{GN}} = \begin{pmatrix} \mathbf{L}_1^\dagger \mathbf{L}_1 & \mathbf{L}_1^\dagger \mathbf{L}_2 & \mathbf{L}_1^\dagger \mathbf{L}_3 \\ \mathbf{L}_2^\dagger \mathbf{L}_1 & \mathbf{L}_2^\dagger \mathbf{L}_2 & \mathbf{L}_2^\dagger \mathbf{L}_3 \\ \mathbf{L}_3^\dagger \mathbf{L}_1 & \mathbf{L}_3^\dagger \mathbf{L}_2 & \mathbf{L}_3^\dagger \mathbf{L}_3 \end{pmatrix}. \quad (32)$$

Each block is a spatially varying *blurring* operator. Its row at  $\mathbf{x}$  is the point-spread function (PSF) of a unit perturbation at  $\mathbf{x}$ , imaged in parameter  $p$ . The diagonal blocks are the familiar

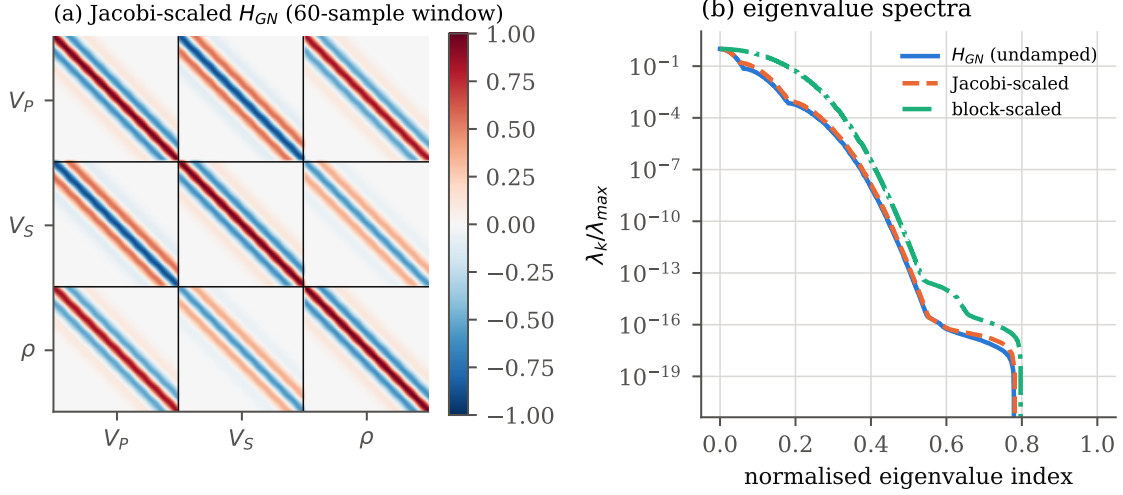


Figure 2: (a) Jacobi-scaled Gauss–Newton Hessian of the linearised elastic AVO toy problem (Section 9, 0–50° incidence) in a 60-sample window. Each of the  $3 \times 3$  blocks is a band whose zero-lag value is the local radiation-pattern correlation. The  $V_P$ – $\rho$  block is nearly as strong as the diagonal blocks (strong positive coupling), and the  $V_P$ – $V_S$  and  $V_S$ – $\rho$  blocks are negative. (b) Eigenvalue spectra of the undamped  $\mathbf{H}_{GN}$ , after Jacobi (diagonal) scaling, and after point-wise  $3 \times 3$  block scaling. The steep fall-off is the band limitation of the wavelet, which no preconditioner can remove. Block scaling lifts the intermediate part of the spectrum, which is the inter-parameter part.

single-parameter PSFs, whose width is set by the bandwidth and aperture. The off-diagonal blocks are *inter-parameter PSFs*: the image in parameter  $p$  of a point scatterer in parameter  $q$ . In a smoothly varying background with good illumination, each block is approximately *local* and *stationary* near  $\mathbf{x}$ . This motivates the local approximations used for preconditioning:

**Diagonal**  $\text{diag}(\mathbf{H}_{GN})$ : one number per parameter and point (illumination and amplitude compensation).

**Block-diagonal (point-wise  $N_p \times N_p$ )  $\mathbf{B}(\mathbf{x})$**  with entries  $\mathbf{B}_{pq}(\mathbf{x}) = \mathbf{H}_{GN}^{(pq)}(\mathbf{x}, \mathbf{x})$ : the zero-lag inter-parameter coupling at each point. Inverting it removes the *local* cross-talk (Section 8.4).

**Banded / PSF-based** Retain a neighbourhood of  $\mathbf{x}$  in each block, which captures spatial deblurring as well (Valenciano et al., 2006; Tang, 2009).

Figure 2 shows these blocks for the toy problem of Section 9. Each of the nine blocks is a banded Toeplitz-like matrix (the wavelet autocorrelation, scaled by an angle-integrated radiation-pattern product). The off-diagonal blocks are as strong as the diagonal ones once each parameter is normalised.

## 5.5 Pseudo-Hessians and illumination

Explicit diagonals of  $\mathbf{H}_{GN}$  require one Born simulation per model point, which is too expensive. Shin et al. (2001) introduced the *pseudo-Hessian*, which keeps the source-side virtual-source energy and neglects the receiver-side Green’s functions:

$$P_p(\mathbf{x}) = \sum_s \int_0^T |(\mathcal{A}_p \mathbf{u}_s)(\mathbf{x}, t)|^2 dt. \quad (33)$$

For  $V_P$  in elasticity,  $\mathcal{A}_{V_P} \mathbf{u} \propto 2\rho V_P \nabla \cdot \mathbf{u}$ , so  $P_{V_P} \propto \int (\nabla \cdot \mathbf{u}_s)^2 dt$ , the *source dilatation energy*. The preconditioned gradient is  $g_p/(P_p + \varepsilon)^\gamma$ , with  $\gamma \in [0.5, 1]$  and a stabilising  $\varepsilon$ . This compensates geometrical spreading and uneven illumination. It *does not* address inter-parameter coupling, because it is diagonal. It may even worsen coupling, because it rescales parameters whose radiation patterns overlap independently (Section 9). A natural multiparameter generalisation is the point-wise  $N_p \times N_p$  *block pseudo-Hessian*  $P_{pq}(\mathbf{x}) = \sum_s \int (\mathcal{A}_p \mathbf{u}_s) \cdot (\mathcal{A}_q \mathbf{u}_s) dt$ , but it contains only source-side information. The angular (two-sided) coupling of Section 6 requires either receiver-side information or the explicit angle-domain construction of Section 8.4.

## 6 Scattering theory and radiation patterns

The Hessian blocks of (32) are expensive to compute but easy to *understand*. In the high-frequency, locally homogeneous limit, every block reduces to an integral over scattering angles of a product of two scalar functions: the radiation patterns. This section derives them and shows that they are exactly the amplitude-versus-angle (AVO) coefficients of exploration seismology.

### 6.1 The Born approximation

Split the model into background and perturbation,  $\mathbf{m} = \mathbf{m}_0 + \delta\mathbf{m}$ , and the wavefield into incident and scattered parts,  $\mathbf{u} = \mathbf{u}_0 + \delta\mathbf{u}$ . To first order in  $\delta\mathbf{m}$ ,

$$\mathcal{A}(\mathbf{m}_0) \delta\mathbf{u} = -\left(\sum_p \mathcal{A}_p \delta m_p\right) \mathbf{u}_0, \quad (34)$$

so the scattered field is radiated by *virtual sources*  $-\mathcal{A}_p \mathbf{u}_0 \delta m_p$  located where the model is perturbed. For a point perturbation at  $\mathbf{x}_0$  and a plane incident wave, the far-field scattered wave has an angular amplitude that depends on the incident direction  $\mathbf{n}$ , the scattered direction  $\mathbf{q}$  and the incident and scattered polarisations  $\mathbf{a}$  and  $\mathbf{g}$ . Appendix B shows that, normalised by  $\rho\omega^2$ ,

$$\mathcal{S} = \frac{\delta\rho}{\rho} (\mathbf{g} \cdot \mathbf{a}) - \frac{1}{\rho c_{\text{in}} c_{\text{out}}} q_j g_i \delta c_{ijkl} n_l a_k, \quad (35)$$

with  $c_{\text{in}}$  and  $c_{\text{out}}$  the incident and scattered wave speeds. The density term is a *force* (dipole) source, projected on the polarisations. The stiffness term is a *moment* source, sensitive to the incident and scattered strain directions. We define the scattering angle  $\theta$  by  $\cos \theta = \mathbf{n} \cdot \mathbf{q}$ . Then  $\theta = 0$  is forward scattering (transmission, diving waves) and  $\theta = \pi$  is backscattering (normal-incidence reflection). For a reflection at incidence angle  $i$ ,  $\theta = \pi - 2i$ .

### 6.2 Acoustic radiation patterns

With  $\mu = 0$  and  $\lambda = \kappa$ , P-to-P scattering ( $\mathbf{a} = \mathbf{n}$ ,  $\mathbf{g} = \mathbf{q}$ ) gives

$$\mathcal{S}_{(\kappa, \rho)} = -\frac{\delta\kappa}{\kappa} + \cos \theta \frac{\delta\rho}{\rho}, \quad (36)$$

$$\mathcal{S}_{(V_P, \rho)} = -2\frac{\delta V_P}{V_P} + (\cos \theta - 1) \frac{\delta\rho}{\rho}, \quad (37)$$

$$\mathcal{S}_{(I_P, \rho)} = -2\frac{\delta I_P}{I_P} + (\cos \theta + 1) \frac{\delta\rho}{\rho}. \quad (38)$$

These three lines already contain most of the parameterization theory of acoustic FWI.

- In  $(\kappa, \rho)$ , the patterns 1 and  $\cos \theta$  are orthogonal over the full aperture  $[0, \pi]$ , but not over a limited one.

- In  $(V_P, \rho)$ , the density pattern vanishes at  $\theta = 0$ . Transmitted and diving waves see *only* velocity, which is why acoustic FWI of diving waves is effectively a velocity inversion. At  $\theta = \pi$  both patterns equal  $-2$  (in units of the relative perturbation): short-offset reflections cannot distinguish  $\delta V_P/V_P$  from  $\delta \rho/\rho$ .
- In  $(I_P, \rho)$ , the density pattern vanishes at  $\theta = \pi$ . Short-offset reflections see *only* impedance. Density no longer leaks into  $I_P$  at normal incidence, but it is resolved only by intermediate and wide angles.

### 6.3 Elastic radiation patterns

For isotropic elasticity, substituting  $\delta c_{ijkl}$  from (14) into (35), with  $\gamma = V_S/V_P$  in the background, gives

$$\mathcal{S}^{PP} = -2 \frac{\delta V_P}{V_P} + 4\gamma^2 \sin^2 \theta \frac{\delta V_S}{V_S} + (\cos \theta - 1 + 2\gamma^2 \sin^2 \theta) \frac{\delta \rho}{\rho} \quad (39)$$

for P-to-P scattering. In the impedance and Lam-Ål parameterizations,

$$\mathcal{S}^{PP} = -2 \frac{\delta I_P}{I_P} + 4\gamma^2 \sin^2 \theta \frac{\delta I_S}{I_S} + (\cos \theta + 1 - 2\gamma^2 \sin^2 \theta) \frac{\delta \rho}{\rho}, \quad (40)$$

$$\mathcal{S}^{PP} = -(1 - 2\gamma^2) \frac{\delta \lambda}{\lambda} - 2\gamma^2 \cos^2 \theta \frac{\delta \mu}{\mu} + \cos \theta \frac{\delta \rho}{\rho}. \quad (41)$$

For P-to-SV conversion ( $\mathbf{a} = \mathbf{n}$ ,  $\mathbf{g} \perp \mathbf{q}$  in the scattering plane), the  $\lambda$  term vanishes identically because  $\mathbf{g} \cdot \mathbf{q} = 0$ , which leaves

$$\mathcal{S}^{PS} = 0 \cdot \frac{\delta V_P}{V_P} + 2\gamma \sin 2\theta \frac{\delta V_S}{V_S} + (-\sin \theta + \gamma \sin 2\theta) \frac{\delta \rho}{\rho}. \quad (42)$$

These closed forms agree with Wu and Aki (1985), Tarantola (1986) and Forgues and Lambaré (1997). The script checks them against a brute-force evaluation of (35) to  $10^{-6}$ . They are plotted in Figures 3 and 4.

### 6.4 Reading the elastic patterns

- **$V_P$  is a monopole** in every parameterization containing it: it scatters equally in all directions and does not convert to S.
- **$V_S$  is invisible in pure transmission and in pure backscattering** for P-P ( $\sin^2 \theta = 0$  at  $0$  and  $\pi$ ). It is best seen at  $\theta \approx 90^\circ$ , i.e. wide-angle reflections around  $45^\circ$  incidence, and in converted waves, which peak at  $\theta \approx 45^\circ$  and  $135^\circ$ .
- **Density is the chameleon.** In the velocity basis it is zero forward and  $-2$  backward, duplicating  $V_P$  at short offsets. In the impedance basis the pattern is  $2$  forward and  $0$  backward. In the Lam-Ål basis it is  $\cos \theta$ , an antisymmetric dipole that is orthogonal to the  $\lambda$  and  $\mu$  patterns over  $[0, \pi]$ , hence the zero correlations in Table 2. For  $\gamma = 0.5$ , the velocity-basis density pattern is  $\cos \theta - 1 + \frac{1}{2} \sin^2 \theta = \mathcal{O}(\theta^4)$  near  $\theta = 0$ : density is extremely weakly sensed by small-angle transmission.
- **Converted waves add independent information.** In (42),  $V_P$  is absent and the  $V_S$  and  $\rho$  patterns have different angular shapes from their P-P counterparts. Adding P-S data breaks the P-P trade-offs (Section 8.9).

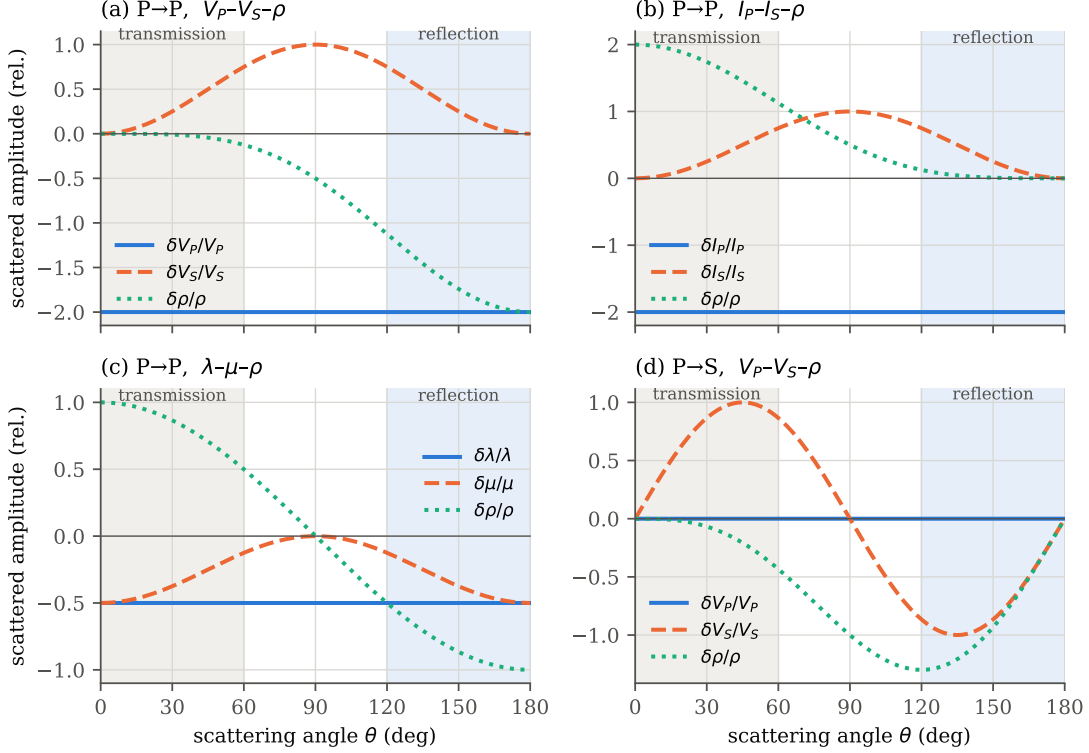


Figure 3: Born radiation patterns (scattered amplitude per unit relative perturbation) versus scattering angle for  $\gamma = V_S/V_P = 0.5$ . (a) P→P in  $(V_P, V_S, \rho)$ , (39). (b) P→P in  $(I_P, I_S, \rho)$ , (40). (c) P→P in  $(\lambda, \mu, \rho)$ , (41). (d) P→S in  $(V_P, V_S, \rho)$ , (42). Grey shading marks the transmission regime and blue shading the reflection regime ( $\theta = \pi - 2i$ ,  $i < 30^\circ$ ). In (a) the  $V_P$  (constant) and  $\rho$  curves meet at  $\theta = 180^\circ$ : normal-incidence reflections cannot separate them. In (b) density vanishes there instead.

## 6.5 Radiation patterns are AVO coefficients

Substituting  $\theta = \pi - 2i$  into (39) and using  $\cos(\pi - 2i) = -\cos 2i$  and  $\sin^2(\pi - 2i) = 4 \sin^2 i \cos^2 i$ , a few lines of algebra give the identity

$$\mathcal{S}^{PP}(\pi - 2i) = -4 \cos^2 i R_{AR}(i), \quad (43)$$

$$R_{AR}(i) = \frac{1}{2}(1 + \tan^2 i) \frac{\delta V_P}{V_P} - 4\gamma^2 \sin^2 i \frac{\delta V_S}{V_S} + \frac{1}{2}(1 - 4\gamma^2 \sin^2 i) \frac{\delta \rho}{\rho}, \quad (44)$$

which is the linearised Aki–Richards reflection coefficient (Aki and Richards, 1980). The script verifies (43) to  $10^{-15}$ . The factor  $\cos^2 i$  is the obliquity (Jacobian) relating a point scatterer to a planar interface. Every AVO result about parameter resolvability therefore transfers directly to reflection-driven FWI. In particular, the Shuey rearrangement

$$R_{AR}(i) = \underbrace{\frac{1}{2} \left( \frac{\delta V_P}{V_P} + \frac{\delta \rho}{\rho} \right)}_{A = \frac{1}{2} \delta I_P / I_P} + \underbrace{\left( \frac{1}{2} \frac{\delta V_P}{V_P} - 4\gamma^2 \frac{\delta V_S}{V_S} - 2\gamma^2 \frac{\delta \rho}{\rho} \right) \sin^2 i}_{B} + \underbrace{\frac{1}{2} \frac{\delta V_P}{V_P}}_C (\tan^2 i - \sin^2 i) \quad (45)$$

shows that, below about  $30^\circ$  incidence where  $\tan^2 i - \sin^2 i \approx \sin^4 i$  is small, P–P reflections constrain essentially *two* combinations (the intercept  $A$  and the gradient  $B$ ) of *three* parameters. Simultaneous  $(V_P, V_S, \rho)$  inversion of short-offset P–P reflections is therefore intrinsically ill-posed. One direction in parameter space, roughly a density/velocity combination that keeps  $A$  and  $B$  fixed, is almost unconstrained. This is the AVO statement of the trade-off quantified in Section 7.

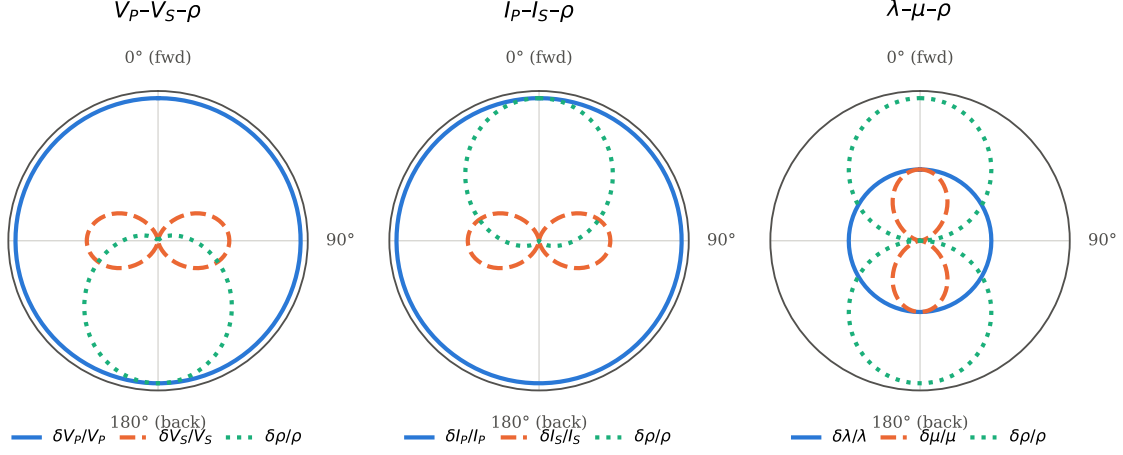


Figure 4: Polar view of  $|S^{PP}(\theta)|$  for the three parameterizations, with forward scattering ( $0^\circ$ ) at the top and backscattering ( $180^\circ$ ) at the bottom. The velocity/impedance patterns are isotropic monopoles. Density changes from a backward lobe (velocity basis) to a forward lobe (impedance basis). Shear parameters radiate at intermediate angles.

## 6.6 Scattering angle and wavenumber

A scattering event with incident and scattered slowness vectors  $\mathbf{n}/c_{\text{in}}$  and  $\mathbf{q}/c_{\text{out}}$  at angular frequency  $\omega$  samples the model wavenumber

$$\mathbf{k} = \omega \left( \frac{\mathbf{n}}{c_{\text{in}}} - \frac{\mathbf{q}}{c_{\text{out}}} \right), \quad |\mathbf{k}|_{PP} = \frac{2\omega}{V_P} \sin \frac{\theta}{2} = \frac{2\omega \cos i}{V_P}. \quad (46)$$

Transmission ( $\theta \rightarrow 0$ ) samples *low* wavenumbers, i.e. the smooth macro-model. Backscattering samples the highest wavenumbers  $2\omega/V_P$ , i.e. the reflectivity (Sirgue and Pratt, 2004). Combining this with the radiation patterns gives a *wavenumber-dependent* picture of resolvability. A parameter whose pattern vanishes in transmission (density in the velocity basis,  $V_S$  for P–P) cannot be recovered at long wavelengths from P–P data at all, whatever the optimiser. Its long-wavelength component lies in the null space. Conversely, the long-wavelength  $V_P$  model is controlled by diving waves, where cross-talk with density is negligible. This is the physical basis of the hierarchical strategies of Section 8.7: long-wavelength  $V_P$  first, from transmissions, then the short-wavelength contrasts of  $V_S$  and  $\rho$ , from wide-angle and converted reflections.

## 6.7 Anisotropic radiation patterns

The same machinery applied to a VTI stiffness gives patterns for  $(V_{P0}, \epsilon, \delta)$  and its reparameterizations (Plessix and Cao, 2011; Gholami et al., 2013; Alkhalifah and Plessix, 2014; Kazei and Alkhalifah, 2019). For acoustic VTI, with the angle  $\phi$  measured from the vertical symmetry axis, a  $\delta$  perturbation scatters mostly at intermediate angles, through the  $\sin^2 \phi \cos^2 \phi$  term of the phase velocity, while  $\epsilon$  scatters at near-horizontal propagation through  $\sin^4 \phi$ . Near-vertical reflections are insensitive to both, which is why  $\delta$  is poorly resolved by surface data. In  $(V_h, \eta, \epsilon)$ -type parameterizations, the well-resolved kinematic parameters are separated from the poorly resolved one, which can then be fixed or tightly regularised.

Table 2: Correlation coefficients  $C_{12}, C_{13}, C_{23}$  of the local P–P Gauss–Newton kernel (47) (uniform angular weight,  $\gamma = 0.5$ ) and the condition number of the correlation matrix, i.e. *after* Jacobi (diagonal) scaling,  $\kappa_J$ . Parameters are numbered in the order given (e.g. 1= $V_P$ , 2= $V_S$ , 3= $\rho$ ). Values from `figures/numbers.txt`.

| Aperture                            | $(V_P, V_S, \rho)$ |          |          |                   | $(I_P, I_S, \rho)$ |          |          |                   | $(\lambda, \mu, \rho)$ |          |          |                   |
|-------------------------------------|--------------------|----------|----------|-------------------|--------------------|----------|----------|-------------------|------------------------|----------|----------|-------------------|
|                                     | $C_{12}$           | $C_{13}$ | $C_{23}$ | $\kappa_J$        | $C_{12}$           | $C_{13}$ | $C_{23}$ | $\kappa_J$        | $C_{12}$               | $C_{13}$ | $C_{23}$ | $\kappa_J$        |
| full, 0–180°                        | −.82               | +.72     | −.49     | 19                | −.82               | −.72     | +.49     | 19                | +.82                   | .00      | .00      | 10                |
| reflection $i \leq 20^\circ$        | −.76               | +.997    | −.71     | $3.7 \times 10^6$ | −.76               | −.61     | +.95     | 103               | +.99                   | +.997    | +.997    | $7.1 \times 10^5$ |
| reflection $i \leq 40^\circ$        | −.80               | +.96     | −.61     | $1.1 \times 10^4$ | −.80               | −.62     | +.92     | 63                | +.86                   | +.94     | +.98     | $1.5 \times 10^3$ |
| reflection $i \leq 60^\circ$        | −.86               | +.87     | −.51     | 334               | −.86               | −.65     | +.83     | 30                | +.74                   | +.66     | +.92     | 38                |
| transmission $\theta \leq 30^\circ$ | −.75               | +.60     | −.95     | 110               | −.75               | −.999    | +.73     | $3.9 \times 10^7$ | +.997                  | −.999    | −.999    | $7.8 \times 10^6$ |
| transmission $\theta \leq 90^\circ$ | −.82               | +.63     | −.91     | 53                | −.82               | −.95     | +.58     | $3.9 \times 10^3$ | +.82                   | −.90     | −.98     | 492               |

## 7 Cross-talk: definition, origin and quantification

### 7.1 The local Hessian kernel

In a locally homogeneous background, the zero-lag Gauss–Newton block at  $\mathbf{x}$  is, up to a spatially varying positive factor (illumination, spreading, bandwidth), an integral of products of radiation patterns over the illuminated scattering angles:

$$\mathbf{B}_{pq}(\mathbf{x}) \propto \int_{\Theta(\mathbf{x})} w(\theta) \mathcal{S}_p(\theta) \mathcal{S}_q(\theta) d\theta \quad (+ \text{sums over wave modes}). \quad (47)$$

Here  $\Theta(\mathbf{x})$  is the set of scattering angles that the acquisition (source and receiver positions, offsets, recording time, mutes) actually produces at  $\mathbf{x}$ , and  $w(\theta)$  is a weight. In 2-D, for example,  $w$  includes the Jacobian from source–receiver coordinates to angles. Equation (47) is the Gram matrix of the radiation patterns on  $\Theta$ . Its normalised off-diagonal entries are the correlation coefficients

$$C_{pq} = \frac{\mathbf{B}_{pq}}{\sqrt{\mathbf{B}_{pp}\mathbf{B}_{qq}}} \in [-1, 1]. \quad (48)$$

$|C_{pq}| \rightarrow 1$  means that the two parameters produce proportional data and cannot be separated locally. The script evaluates (47) with  $w = 1$  for the reflection apertures  $\Theta = [\pi - 2i_{\max}, \pi]$  and the transmission apertures  $\Theta = [0, \theta_{\max}]$ . Table 2 and Figure 5 summarise the results.

### 7.2 What the numbers say

1. **Cross-talk is the norm, not the exception.** Even with *full* aperture (0–180°, which no surface survey achieves),  $|C_{V_P V_S}| = 0.82$ . The gradient of a velocity-basis inversion always leaks between parameters.
2. **Short-offset reflection is the worst case for  $(V_P, \rho)$ .** At  $i \leq 20^\circ$ ,  $C_{V_P \rho} = 0.997$  and  $\kappa_J = 3.7 \times 10^6$ . The radiation patterns are almost identical and the data constrain only  $I_P$ . Figure 6 shows the acoustic analogue: the misfit valley in  $(\delta V_P/V_P, \delta \rho/\rho)$  is a long diagonal trough along the impedance-preserving direction  $(1, -1)$ .
3. **Reparameterization changes which pair is coupled, not whether coupling exists.** In  $(I_P, I_S, \rho)$ , the worst pair at short offsets is  $I_S - \rho$  (0.95), but the condition number is  $10^4$  times smaller, and the poorly resolved direction now lies mainly along  $(I_S, \rho)$ , leaving  $I_P$  clean (Table 3).
4. **The best parameterization depends on the data regime.** For transmission, the velocity basis is best ( $\kappa_J = 110$  versus  $3.9 \times 10^7$ ). For reflection, the impedance basis is

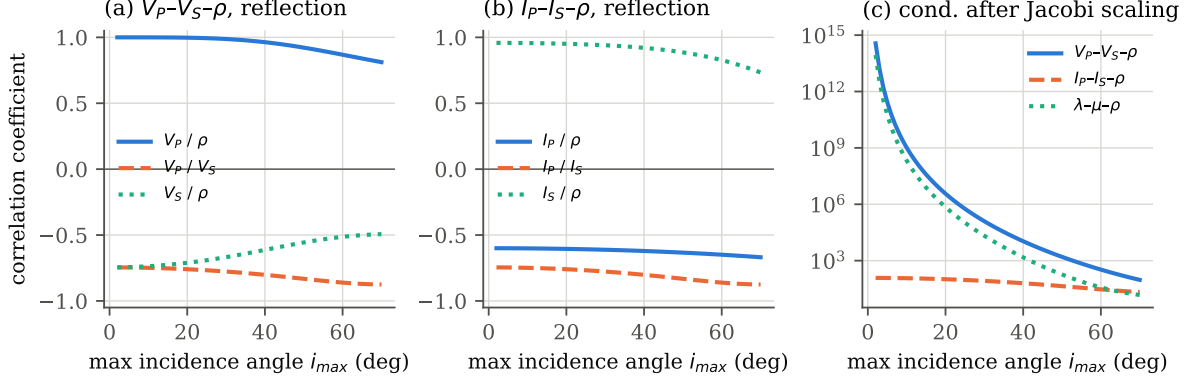


Figure 5: Cross-talk of the local P-P Gauss-Newton kernel versus the maximum reflection incidence angle. (a) Correlation coefficients in  $(V_P, V_S, \rho)$ :  $V_P$ - $\rho$  coupling is almost perfect ( $> 0.99$ ) below  $25^\circ$ . (b) In  $(I_P, I_S, \rho)$  the dominant coupling moves to  $I_S$ - $\rho$  and the  $I_P$  coupling falls to about 0.6. (c) Condition number after Jacobi scaling, which is the part of the ill-conditioning that diagonal preconditioning *cannot* remove. The impedance basis is better conditioned by four orders of magnitude at short offsets.

Table 3: Eigen-decomposition of the Jacobi-scaled local kernel for reflections with  $i \leq 40^\circ$  (P-P only). The weakest eigenvector is the combination the data barely see.

|                    | $\lambda_{\min}$     | weakest eigenvector     | interpretation                              |
|--------------------|----------------------|-------------------------|---------------------------------------------|
| $(V_P, V_S, \rho)$ | $2.4 \times 10^{-4}$ | $(+0.77, +0.26, -0.58)$ | $V_P$ up, $\rho$ down: impedance-preserving |
| $(I_P, I_S, \rho)$ | $4.1 \times 10^{-2}$ | $(-0.29, -0.78, +0.56)$ | mostly $I_S$ versus $\rho$                  |

best. The Lam-År basis is excellent only with full aperture, where density decouples exactly ( $C_{\lambda\rho} = C_{\mu\rho} = 0$ ), and it is poor otherwise.

5. **Aperture is the master variable.** Every coupling weakens as  $i_{\max}$  grows (Figure 5c). Long offsets and wide azimuths are the most effective “cross-talk remedy” of all, but they are an acquisition decision.

### 7.3 Sensitivity imbalance versus coupling

Two different things make multiparameter Hessians ill-conditioned, and they need different treatments.

**Sensitivity (scale) imbalance** The diagonal entries  $\mathbf{B}_{pp}$  differ by orders of magnitude. This happens because units differ ( $V_P \sim 3000 \text{ m/s}$  against  $\rho \sim 2 \text{ g/cm}^3$  gives a factor of  $10^6$  in  $\mathbf{H}$  before any physics), because radiation amplitudes differ ( $|\mathcal{S}_{V_P}| = 2$  while  $|\mathcal{S}_{V_S}| \leq 4\gamma^2 = 1$ ), and because illumination differs. Imbalance is cured by scaling, i.e. *diagonal* preconditioning.

**Coupling** Large  $|C_{pq}|$ . The correlation matrix itself is ill-conditioned (large  $\kappa_J$ ). Diagonal scaling leaves it untouched by construction. Only off-diagonal (block, quasi-Newton or Newton) information, or a better parameterization, can reduce it.

In the transmission row of Table 2, the raw condition number in  $(V_P, V_S, \rho)$  is  $7.6 \times 10^6$  but  $\kappa_J$  is only 110. Almost all of that ill-conditioning is imbalance:  $V_S$  and  $\rho$  are simply barely sensed. In the reflection rows, by contrast,  $\kappa_J$  is itself huge, and the problem is coupling.

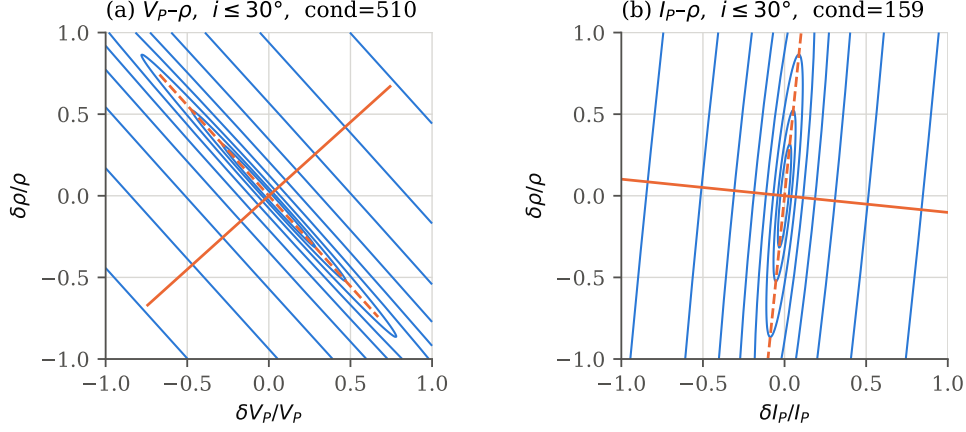


Figure 6: Quadratic misfit contours (blue) of the acoustic two-parameter problem for reflections with  $i \leq 30^\circ$  and the Hessian eigenvectors (orange; dashed = weakest). (a)  $(V_P, \rho)$ : correlation 0.996, with an elongated valley along the impedance-preserving diagonal. Steepest descent zig-zags across it and the update leaks between parameters. (b)  $(I_P, \rho)$ : the eigenvectors are almost aligned with the axes (to within  $6^\circ$ ).  $I_P$  is well determined, and the ambiguity is confined to density.

#### 7.4 Coupling, trade-off and posterior correlation

For two parameters with a  $2 \times 2$  local Hessian  $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$ , the posterior covariance (4) is  $\frac{1}{ab-c^2} \begin{pmatrix} b & -c \\ -c & a \end{pmatrix}$ . The posterior correlation is therefore  $-C_{12}$ . *Positive Hessian coupling means negatively correlated posterior uncertainty* (and vice versa), and the posterior variances grow as  $1/(1 - C_{12}^2)$ . With  $C_{12} = 0.997$ , the uncertainty along the weak direction is  $\sim 13$  times larger than without coupling. The data determine  $\delta V_P/V_P + \delta \rho/\rho$  well and  $\delta V_P/V_P - \delta \rho/\rho$  hardly at all. This is the trade-off. Any estimate of the individual parameters is then dominated by noise, regularisation or the starting model (Fichtner and Trampert, 2011b).

#### 7.5 Leakage from modelling errors

The Hessian analysis assumes the simulator is correct. When it is not, the residual contains a systematic component  $\mathbf{r}_{\text{err}}$  that is not in the range of  $\mathbf{L}$ . Its migration  $\mathbf{L}^\dagger \mathbf{r}_{\text{err}}$  projects onto the parameters with the most similar radiation patterns. Typical cases are the following:

- **Ignoring density** in acoustic FWI: reflection amplitudes due to  $\delta \rho$  are explained by  $V_P$  contrasts, which overshoot impedance contrasts in velocity.
- **Ignoring attenuation**: amplitude decay with offset and time is explained by velocity and density, and dispersion by velocity (Kamei and Pratt, 2013; Keating and Innanen, 2019).
- **Ignoring anisotropy**: the mismatch between vertical and horizontal velocities causes mis-positioned reflectors and smeared velocity updates.
- **Acoustic approximation of elastic data**: AVO effects from  $V_S$  contrasts and converted-wave energy appear as artefacts in  $V_P$  and  $\rho$ .
- **Wavelet and amplitude errors** (source directivity, receiver coupling, incorrect geometrical spreading in 2-D): these are absorbed preferentially by the parameters with the largest amplitude sensitivity, usually density and impedances.

No preconditioner can remove this leakage. It requires better physics, or data selection and weighting that suppress the unmodelled part (Section 8.8).

## 7.6 Measuring cross-talk in practice

1. **Local radiation-pattern analysis:** evaluate (47) with the survey’s actual angle coverage per depth. This is cheap and should precede any multiparameter project.
2. **Inter-parameter point-spread functions:** apply  $\mathbf{H}$  (via Hessian-vector products) to a comb of point perturbations in parameter  $q$  only and inspect the response in every  $p$  (Fichtner and Trampert, 2011b; Pan et al., 2018). Random probing estimates diagonal and block-diagonal entries stochastically (Fichtner and van Leeuwen, 2015; Trampert et al., 2013).
3. **Single-parameter synthetic tests:** invert synthetic data generated with an anomaly in only one parameter and measure the energy that appears in the others. Staggered checkerboards, with anomalies placed at different positions for different parameters, make the leakage visible (Köhn et al., 2012; Operto et al., 2013).
4. **Cross-talk index:** the fraction of update energy in the “wrong” parameters around single-parameter anomalies (used in Section 9).
5. **Eigen-analysis** of small Hessians (local, or restricted to a target) identifies the weakly constrained combinations directly (Table 3).

## 8 Techniques for resolving cross-talk

The remedies fall into four families, which correspond to the factors in  $\mathbf{H}\delta\mathbf{m} = -\mathbf{g}$ : (i) change the *unknowns* (parameterization and scaling), (ii) approximate  $\mathbf{H}^{-1}$  better (preconditioning, quasi-Newton, truncated Newton), (iii) change the *data* that enter  $\mathbf{L}$  (selection, weighting, hierarchy, multicomponent), (iv) add *information* (regularisation, priors, constraints). Family (ii) attacks coupling. Families (iii) and (iv) are the only ones that can do anything about trade-off. Family (i) helps with both.

### 8.1 Choosing the parameterization

By (17), a reparameterization changes the Hessian by congruence. The selection criteria, in order of importance, are:

1. **Make the well-resolved parameters orthogonal to the poorly resolved ones** over the apertures that dominate the data. For reflection-dominated data,  $(I_P, I_S, \rho)$  isolates density (Figure 6b). For transmission-dominated data,  $(V_P, V_S, \rho)$  isolates density, because its pattern vanishes forward.
2. **Minimise  $\kappa_J$**  of the local kernel over the expected angle coverage (Table 2). This can be computed in minutes before any inversion (Köhn et al., 2012; Gholami et al., 2013; Pan et al., 2018).
3. **Match the scale content.** Parameters controlling kinematics (velocities) need low wavenumbers, which come from transmissions. Parameters controlling only amplitudes (density, some anisotropy parameters) are recoverable only at high wavenumbers (Section 6.6).
4. **Match the deliverable.** If the goal is  $V_P/V_S$  or Poisson’s ratio, invert for it directly. Its posterior uncertainty is then not inflated by propagating correlated errors.

Published conclusions broadly agree. For elastic FWI with surface data,  $(I_P, I_S, \rho)$  or  $(V_P, V_S, \rho)$  with density updated last or fixed is preferred for the velocities (Tarantola, 1986; Köhn et al., 2012). For acoustic VTI, parameterizations that separate a dominant velocity ( $V_h$  or  $V_n$ ) from  $\epsilon$  and  $\delta$ , with  $\delta$  treated as poorly resolved, are preferred (Plessix and Cao, 2011; Gholami et al., 2013; Alkhalifah and Plessix, 2014). The companion SymPy study of this project ranks the seven parameterizations implemented in the production code by  $\kappa(\mathbf{H}_{\text{GN}})$  and reaches consistent conclusions: impedance-type bases for reflection, velocity-type bases for transmission.

## 8.2 Scaling and normalisation

A diagonal change of variables  $\mathbf{m} = \mathbf{D}\tilde{\mathbf{m}}$  is the same as diagonal preconditioning:

$$\tilde{\mathbf{g}} = \mathbf{D}\mathbf{g}, \quad \tilde{\mathbf{H}} = \mathbf{D}\mathbf{H}\mathbf{D}, \quad -\tilde{\mathbf{g}} \text{ in } \tilde{\mathbf{m}} \iff -\mathbf{D}^2\mathbf{g} \text{ in } \mathbf{m}. \quad (49)$$

Choose  $\mathbf{D}$  so that all parameter classes are dimensionless and of order one, for example  $\tilde{m}_p = (m_p - m_{p,0})/m_{p,\text{ref}}$  or  $\tilde{m}_p = \ln(m_p/m_{p,\text{ref}})$ . Better still, equalise the diagonal of  $\tilde{\mathbf{H}}$  class by class. Without scaling, the first steepest-descent step is dominated by the parameter class with the largest raw gradient, and the line search, which chooses a single  $\alpha$ , is meaningful only for that class. Quasi-Newton methods are *not* scale-invariant at start-up, because their initial matrix is  $\gamma\mathbf{I}$ , and limited-memory updates correct a bad scaling only slowly. Newton’s method is affine-invariant in exact arithmetic, but its inner CG solver is not, so scaling still matters as a preconditioner.

### Scaling checklist

- Normalise every class to  $\mathcal{O}(1)$  before inversion. Logarithmic parameters are a robust default.
- Verify the balance: after the first gradient, the RMS of each preconditioned gradient class, relative to its model RMS, should be comparable.
- Never form joint vector norms (for step lengths, finite-difference steps or convergence tests) over unscaled mixed-unit vectors. See Section 10.2 for a real failure mode.

## 8.3 Diagonal preconditioning and its limits

Diagonal preconditioners (pseudo-Hessian (33), illumination compensation, depth gain, the exact  $\text{diag}(\mathbf{H}_{\text{GN}})$  estimated by probing) are standard and indispensable for balancing illumination in space. Applied *per parameter class*, they also cure sensitivity imbalance. They cannot reduce coupling, and they can make it *worse*. If parameter  $q$  is weakly sensed ( $\mathbf{B}_{qq}$  small) but coupled to a strongly sensed parameter  $p$ , dividing  $g_q$  by  $\mathbf{B}_{qq}$  amplifies the leakage of  $p$  into  $q$ . The laboratory of Section 9 measures exactly this effect: Jacobi scaling raises the cross-talk index from 0.41 to 0.75, because it amplifies the leakage into the weakly sensed  $V_S$ . For a  $20^\circ$  aperture, the single-scatterer tests show a  $V_P$ -only anomaly producing a  $V_S$  response 7.3 times larger than the  $V_P$  response after diagonal scaling (Table 5).

## 8.4 Block-diagonal (point-wise $N_p \times N_p$ ) preconditioning

The most cost-effective direct attack on coupling is to invert, at every model point, the local  $N_p \times N_p$  block  $\mathbf{B}(\mathbf{x})$  of (32):

$$\mathbf{s}(\mathbf{x}) = -(\mathbf{B}(\mathbf{x}) + \mu(\mathbf{x})\mathbf{D}(\mathbf{x}))^{-1}\mathbf{g}(\mathbf{x}), \quad (50)$$

where  $\mathbf{D}$  is  $\text{diag}(\mathbf{B})$  and  $\mu$  a small relative damping. The cost is that of solving one  $3 \times 3$  system per grid point. Ways to obtain  $\mathbf{B}(\mathbf{x})$ :

1. **Analytic angle-domain construction.** Estimate the local scattering-angle coverage  $\Theta(\mathbf{x})$ , for example by ray tracing or by the Poynting vectors of the source and receiver wavefields during the gradient computation. Then evaluate (47) with the radiation patterns of Section 6, multiplied by the diagonal pseudo-Hessian amplitude.
2. **Hessian probing.** Apply  $\mathbf{H}$  (by second-order adjoint or finite difference) to  $N_p$  sparse combs of point perturbations, one per parameter class, and read off the  $N_p \times N_p$  responses at the comb points. Interpolate between them. Random-sign probing vectors give unbiased estimates of the diagonal blocks (Fichtner and van Leeuwen, 2015).
3. **Extended imaging.** Angle-domain common-image gathers from the gradient computation provide per-angle images whose AVO behaviour can be inverted point-wise with (44). This is the classical AVO inversion, which here acts as a preconditioner.

*Why damping matters.* Where  $\mathbf{B}(\mathbf{x})$  is nearly singular (trade-off), its exact inverse amplifies noise and modelling error enormously along the weak eigenvector. The damping  $\mu$  (or a truncated eigen-inverse of  $\mathbf{B}$ ) keeps the update finite there and reverts gracefully to diagonal scaling. In the laboratory (Section 9), one application of the damped  $3 \times 3$  block inverse reduces the cross-talk index of the gradient from 0.41 to 0.06. As a preconditioner for CG it is the fastest method tested.

## 8.5 Quasi-Newton methods

l-BFGS (Nocedal, 1980; Liu and Nocedal, 1989) approximates the inverse Hessian from gradient differences. Each pair  $(\mathbf{s}_j, \mathbf{y}_j) = (\mathbf{m}_{j+1} - \mathbf{m}_j, \mathbf{g}_{j+1} - \mathbf{g}_j)$  supplies one exact curvature condition,  $\mathbf{B}_k \mathbf{s}_j = \mathbf{y}_j$ , which includes the inter-parameter part. l-BFGS therefore removes coupling *gradually* along the directions it has explored. It is the workhorse of multiparameter FWI (Brossier et al., 2009; Prieux et al., 2013a) and is at its best when

- the parameters are well scaled (it cannot easily fix scaling),
- it is combined with a good diagonal or block preconditioner as the initial matrix  $\mathbf{B}_0^{-1}$  (“preconditioned l-BFGS”),
- the memory  $\ell$  (typically 5–20) captures the dominant coupling modes, which are few in number because there are  $N_p$  classes, although the spatial dimension is large.

Its limitation is the slow start: in the first few iterations the update is nearly the (pre-conditioned) gradient and carries its full cross-talk. Since multiparameter FWI runs typically use few iterations per frequency band, the early updates matter.

## 8.6 Truncated Newton (Newton-CG)

Truncated Newton solves the Newton system (9) *approximately* with an inner conjugate-gradient loop. CG needs only Hessian-vector products. The inner solution is the Krylov-subspace approximation of  $-\mathbf{H}^{-1}\mathbf{g}$ , which, after a few inner iterations, already removes the dominant coupling (Métivier et al., 2013, 2014, 2017). Algorithm 1 gives the version implemented in `tdfwi-hess/hessian.py` (function `newton_cg`).

### Design choices.

- **Gauss–Newton or full Newton.** The GN product (29) is always SPD, so CG is well defined. The full product (second-order adjoint or finite difference) includes  $\mathbf{Q}$ , which carries double-scattering information and improves convergence when multiples matter (Métivier et al., 2014), but it can be indefinite (Figure 1c). Hence the negative-curvature

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**Algorithm 1** Preconditioned truncated-Newton (Newton-CG) search direction

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**Require:** gradient  $\mathbf{g}$ , Hessian-vector product  $\mathbf{v} \mapsto \mathbf{H}\mathbf{v}$ , preconditioner  $\mathbf{M}^{-1} \approx \mathbf{H}^{-1}$  (SPD), maximum inner iterations  $n_{cg}$ , forcing term  $\eta_k$

- 1:  $\mathbf{s} \leftarrow \mathbf{0}$ ;  $\mathbf{r} \leftarrow -\mathbf{g}$ ;  $\mathbf{z} \leftarrow \mathbf{M}^{-1}\mathbf{r}$ ;  $\mathbf{p} \leftarrow \mathbf{z}$
- 2: **for**  $j = 1, \dots, n_{cg}$  **do**
- 3:    $\mathbf{h} \leftarrow \mathbf{H}\mathbf{p}$  {one Hessian-vector product = two wave simulations per source}
- 4:   **if**  $\mathbf{p}^\top \mathbf{h} \leq \epsilon_c \|\mathbf{p}\| \|\mathbf{h}\|$  **then**
- 5:     **if**  $j = 1$  **then**  $\mathbf{s} \leftarrow -\mathbf{M}^{-1}\mathbf{g}$  **end if**; **break** {negative / zero curvature}
- 6:   **end if**
- 7:    $\alpha \leftarrow \mathbf{r}^\top \mathbf{z} / \mathbf{p}^\top \mathbf{h}$ ;    $\mathbf{s} \leftarrow \mathbf{s} + \alpha \mathbf{p}$ ;    $\mathbf{r} \leftarrow \mathbf{r} - \alpha \mathbf{h}$
- 8:   **if**  $\|\mathbf{r}\| \leq \eta_k \|\mathbf{g}\|$  **then**
- 9:     **break** {Eisenstat–Walker forcing}
- 10:   **end if**
- 11:    $\mathbf{z}_{\text{new}} \leftarrow \mathbf{M}^{-1}\mathbf{r}$ ;    $\beta \leftarrow \mathbf{r}^\top \mathbf{z}_{\text{new}} / \mathbf{r}_{\text{old}}^\top \mathbf{z}_{\text{old}}$ ;    $\mathbf{p} \leftarrow \mathbf{z}_{\text{new}} + \beta \mathbf{p}$
- 12: **end for**
- 13: **if**  $\mathbf{g}^\top \mathbf{s} \geq 0$  **then**
- 14:    $\mathbf{s} \leftarrow -\mathbf{M}^{-1}\mathbf{g}$  {descent safeguard}
- 15: **end if**
- 16: **return**  $\mathbf{s}$  (followed by an outer line search on  $J(\mathbf{m} + \alpha \mathbf{s})$ )

---

test in line 4. The test must be *relative*: physical-unit FWI curvatures are tiny in absolute terms, and an absolute threshold would trigger spuriously.

- **Forcing terms.** Solving the inner system accurately far from the solution is wasteful. The Eisenstat–Walker choices (Eisenstat and Walker, 1996), e.g.  $\eta_k = \min(0.5, \|\mathbf{g}_k\| - \|\mathbf{g}_{k-1} + \mathbf{H}_{k-1}\mathbf{s}_{k-1}\| / \|\mathbf{g}_{k-1}\|)$ , tighten the tolerance as the outer iterations converge.
- **Preconditioning.** The inner CG converges at a rate governed by  $\text{cond}(\mathbf{M}^{-1}\mathbf{H})$ . The diagonal pseudo-Hessian balances illumination. A block preconditioner (Section 8.4) also removes local coupling, so that the expensive inner iterations are spent on the non-local (spatial) part of  $\mathbf{H}$ . An l-BFGS approximation can serve as  $\mathbf{M}^{-1}$  as well (Métivier et al., 2013).
- **Cost.** Each outer iteration costs one gradient plus  $n_{cg}$  Hessian-vector products, i.e.  $2(1 + n_{cg})$  simulations per source (forward-difference or second-order adjoint) or  $2(1 + 2n_{cg})$  (central difference). With  $n_{cg} = 3\text{--}8$ , one TN iteration costs as much as 4–9 l-BFGS iterations. It pays off when it reaches a better, less cross-talk-contaminated model in fewer outer iterations. Source subsampling in the inner loop reduces the cost substantially (Matharu and Sacchi, 2019).

## 8.7 Hierarchical and multi-stage strategies

Because resolvability depends on wavenumber and wave type (Section 6.6), multiparameter inversions are almost always staged (Tarantola, 1986; Prioux et al., 2013a; Operto et al., 2013):

1. **Frequency continuation** (Bunks et al., 1995; Sirgue and Pratt, 2004): low to high frequencies. This mitigates cycle skipping, and it orders the wavenumbers so that the macro-model (mostly  $V_P$ ) converges first.
2. **Parameter staging**: invert the best-resolved class first (e.g.  $V_P$ , or  $I_P$ ), possibly with the others tied by a rock-physics relation. Then release the second class, and finally density, with fewer iterations and stronger regularisation. Staging is a hard form of trade-off control. It is effective but biased if the frozen parameters are wrong.

3. **Simultaneous inversion with staged data** is often better than staging the parameters: invert all classes jointly, but grow the data from early arrivals (transmission,  $V_P$ -dominated) to full offsets and later arrivals (reflections, converted waves). The Hessian then remains well conditioned at every stage.

## 8.8 Data selection and weighting

Data weighting  $\mathbf{W}_d$  reshapes the angle weight  $w(\theta)$  in (47) and therefore the Hessian. Useful tools are:

- **Offset or angle weighting:** up-weighting long offsets reduces  $V_P$ - $\rho$  coupling (Figure 5). Down-weighting near offsets limits the influence of density-dominated amplitudes when density is not a target.
- **Time windows and mutes:** separating diving waves from reflections lets the inversion assign each data class to the parameters it resolves.
- **Amplitude-insensitive misfits** (phase-only, normalised cross-correlation, traveltimes) for the kinematic stage. These suppress leakage from amplitude modelling errors into impedance and density.
- **Mode separation:** P and S potentials or their surrogates (divergence and curl of multi-component data) with separate weights.

## 8.9 Multicomponent data and converted waves

Converted P-S waves have radiation patterns (42) that are *linearly independent* of the P-P ones. In particular,  $V_P$  does not convert at all. Adding them to the local kernel changes the reflection,  $i \leq 40^\circ$ ,  $(V_P, V_S, \rho)$  results (script output) from

$$\begin{aligned} \text{P-P only: } & C_{V_P\rho} = 0.96, \quad C_{V_S\rho} = -0.61, \quad \kappa_J = 1.1 \times 10^4, \\ \text{P-P + P-S: } & C_{V_P\rho} = 0.80, \quad C_{V_S\rho} = +0.13, \quad \kappa_J = 256, \end{aligned}$$

a forty-fold improvement in conditioning. Multicomponent ocean-bottom data are therefore the data type of choice for elastic multiparameter FWI (Sears et al., 2008; Prioux et al., 2013b). They demand an elastic simulator, accurate receiver coupling and orientation, and careful handling of interface (Scholte) waves.

## 8.10 Regularisation, priors and constraints

When the data leave a combination unresolved, information must come from elsewhere:

- **Tikhonov / damping** ((11)) with class-specific weights: penalising the poorly resolved class more heavily is equivalent to a prior with smaller variance for it.
- **Model covariance preconditioning:** smoothing operators applied to the gradient per class, with longer correlation lengths for the poorly resolved classes, encode the scale dependence of Section 6.6.
- **Structural coupling:** cross-gradient (Gallardo and Meju, 2003) or joint total-variation penalties favour co-located boundaries between classes without imposing values. This is valuable when the classes share geology but have different sensitivities.
- **Rock-physics relations:** Gardner's  $\rho = 0.31 V_P^{0.25}$  ( $\rho$  in g/cm<sup>3</sup>,  $V_P$  in m/s) (Gardner et al., 1974), mudrock lines for  $V_S(V_P)$ , or facies-dependent trends, used as soft priors  $\|\rho - f(V_P)\|^2$ . Alternatively, reparameterize as  $\rho = f(V_P) + \delta\rho$  and regularise only  $\delta\rho$ .

- **Bounds and physical constraints:**  $V_S < V_P/\sqrt{2}$  (positive  $\lambda$ ), positivity, plausible  $V_P/V_S$  ranges. Projections onto convex sets enforce them without destroying convergence (Baumstein, 2013).

## 8.11 Model-domain deblurring and subspace methods

Two further ideas complete the toolbox:

- **Non-stationary matching filters** (Guitton, 2004; Aoki and Schuster, 2009): estimate a filter that maps  $\mathbf{H}\mathbf{v}$  back to  $\mathbf{v}$  for a known  $\mathbf{v}$  (e.g. the current gradient), and apply it to the gradient. The multiparameter version maps an  $N_p$ -vector of images to an  $N_p$ -vector, i.e. a spatially local block filter. It approximates  $\mathbf{H}^{-1}$ , including the coupling, at the cost of a single Hessian-vector product.
- **Subspace methods** (Kennett et al., 1988): restrict each update to the span of a few basis vectors per parameter class (e.g. the preconditioned gradient of each class and their cross-terms), compute the small projected Hessian exactly with Hessian-vector products, and solve it. This is an exact, small-dimensional treatment of inter-class coupling. It was designed precisely for problems with “multiple parameter classes”.

## 8.12 Summary of remedies

Table 4: Remedies for multiparameter cross-talk.

| Remedy                               | Coupling              | Trade-off | Extra cost               | Main caveat                              |
|--------------------------------------|-----------------------|-----------|--------------------------|------------------------------------------|
| Reparameterization                   | reduces               | confines  | none                     | depends on data regime                   |
| Scaling / normalisation              | — (imbalance)         | —         | none                     | mandatory, not sufficient                |
| Diagonal precondition.               | no (may worsen)       | no        | low                      | balances illumination only               |
| Block $N_p \times N_p$ precondition. | local                 | no        | low–medium               | needs angle coverage or probing; damp it |
| l-BFGS                               | gradual               | no        | none                     | slow start, needs scaling                |
| Truncated Newton                     | yes (global)          | no        | $n_{cg} \times$ gradient | cost; indefinite $\mathbf{H}$            |
| Hierarchy / staging                  | indirect              | partially | time                     | bias from frozen classes                 |
| Data weighting                       | reshapes $\mathbf{H}$ | partially | none                     | discards information                     |
| Multicomponent data                  | yes                   | yes       | acquisition              | elastic modelling needed                 |
| Regularisation / priors              | no                    | yes       | low                      | bias towards prior                       |

# 9 A numerical laboratory

This section makes the preceding theory concrete with a problem small enough to form every matrix explicitly yet rich enough to show all three faces of cross-talk. Everything here is computed by `make_figures.py`.

## 9.1 Set-up: linearised elastic AVO as Born inversion

Following Section 6.5, a horizontally layered elastic medium with background  $\gamma = 0.5$  is represented by three *relative-perturbation traces*  $\mathbf{a}(t) = \delta V_P/V_P$ ,  $\mathbf{b}(t) = \delta V_S/V_S$  and  $\mathbf{c}(t) = \delta \rho/\rho$

in two-way time. These are 300 samples at 2 ms, i.e. 900 unknowns. For each incidence angle  $i$  ( $0\text{--}50^\circ$  in  $1^\circ$  steps, 51 angle traces) the data are

$$\mathbf{d}(t, i) = w(t) * \left[ \frac{1}{2}(1 + \tan^2 i) \mathbf{a} - 4\gamma^2 \sin^2 i \mathbf{b} + \frac{1}{2}(1 - 4\gamma^2 \sin^2 i) \mathbf{c} \right](t) + \mathbf{n}, \quad (51)$$

with a 30 Hz Ricker wavelet  $w$  and Gaussian noise  $\mathbf{n}$  at 2% of the peak amplitude. This is the Born operator  $\mathbf{L} = [\mathbf{L}_{V_P} \ \mathbf{L}_{V_S} \ \mathbf{L}_\rho]$  for scatterers seen at reflection angles. It is linear, so  $\mathbf{H}_{\text{GN}} = \mathbf{L}^\top \mathbf{L}$  is the exact Hessian. The true model has four scatterers (Figure 7, left): a  $V_P$ -only perturbation at 0.10 s, a  $V_S$ -only perturbation at 0.22 s, a  $\rho$ -only perturbation at 0.34 s, and a “gas sand” at 0.47 s with  $(\delta V_P/V_P, \delta V_S/V_S, \delta \rho/\rho) = (-0.08, +0.05, -0.06)$ . The band-limited wavelet leaves  $\mathbf{H}_{\text{GN}}$  singular to working precision ( $\kappa = 1.4 \times 10^{19}$ : frequencies outside the wavelet band are a null space). We therefore solve the damped system  $(\mathbf{H}_{\text{GN}} + \mu \mathbf{I})\delta \mathbf{m} = \mathbf{L}^\top \mathbf{d}$  with  $\mu = 10^{-3} \max \text{diag} \mathbf{H}_{\text{GN}}$ , whose condition number is  $1.3 \times 10^4$ . Its solution  $\delta \mathbf{m}^*$  is the best any optimiser can achieve with these data and this damping. We compare methods by their distance to  $\delta \mathbf{m}^*$  and by the *cross-talk index* (CTI): the fraction of model energy in the two “wrong” parameters within  $\pm 12$  ms of the three single-parameter scatterers (CTI = 0 for the true model).

## 9.2 Single-scatterer leakage

First consider a single point scatterer with no wavelet blur, i.e. the zero-lag  $3 \times 3$  block  $\mathbf{B}$ . Its gradient response is a column of  $\mathbf{B}$ , its Jacobi-preconditioned response is that column divided by  $\text{diag} \mathbf{B}$ , and its block-preconditioned response is  $\mathbf{B}^{-1} \mathbf{B} \mathbf{e}_p = \mathbf{e}_p$ . Table 5 lists the responses.

Table 5: Response, at the scatterer location, to a unit relative perturbation of one parameter, normalised by the response of the true parameter, for three reflection apertures. “Block” is exactly  $\mathbf{e}_p$  by construction, i.e. no local leakage. Note the large *Jacobi* leakage into the weakly sensed  $V_S$ .

| $i_{\text{max}}$ | true   | gradient ( $\mathbf{L}^\top \mathbf{L} \mathbf{e}_p$ ) |       |        | Jacobi ( $\mathbf{D}^{-1} \mathbf{L}^\top \mathbf{L} \mathbf{e}_p$ ) |       |        |
|------------------|--------|--------------------------------------------------------|-------|--------|----------------------------------------------------------------------|-------|--------|
|                  |        | $V_P$                                                  | $V_S$ | $\rho$ | $V_P$                                                                | $V_S$ | $\rho$ |
| 20°              | $V_P$  | 1                                                      | −0.08 | 0.92   | 1                                                                    | −7.3  | 1.08   |
|                  | $V_S$  | −7.3                                                   | 1     | −6.25  | −0.08                                                                | 1     | −0.08  |
|                  | $\rho$ | 1.08                                                   | −0.08 | 1      | 0.92                                                                 | −6.25 | 1      |
| 40°              | $V_P$  | 1                                                      | −0.28 | 0.67   | 1                                                                    | −2.65 | 1.35   |
|                  | $V_S$  | −2.65                                                  | 1     | −1.42  | −0.28                                                                | 1     | −0.30  |
|                  | $\rho$ | 1.35                                                   | −0.30 | 1      | 0.67                                                                 | −1.42 | 1      |
| 60°              | $V_P$  | 1                                                      | −0.39 | 0.29   | 1                                                                    | −2.31 | 1.80   |
|                  | $V_S$  | −2.31                                                  | 1     | −0.52  | −0.39                                                                | 1     | −0.54  |
|                  | $\rho$ | 1.80                                                   | −0.54 | 1      | 0.29                                                                 | −0.52 | 1      |

Three lessons emerge: (1) In the raw gradient, a pure  $V_S$  anomaly appears *more strongly in  $V_P$  and  $\rho$  than in  $V_S$  itself* (−7.3 and −6.25 at 20°), because the  $V_S$  radiation pattern is weak at short offsets while its coupling to the others is large. (2) Jacobi scaling simply transposes the pattern. It cleans the  $V_S$  column but contaminates  $V_S$  with  $V_P$  and  $\rho$  leakage. (3) Only the off-diagonal information in  $\mathbf{B}^{-1}$  removes the leakage, and the leakage shrinks with aperture in every column.

## 9.3 Updates and convergence with a band-limited wavelet

Figures 7 and 8 and Table 6 summarise the results.

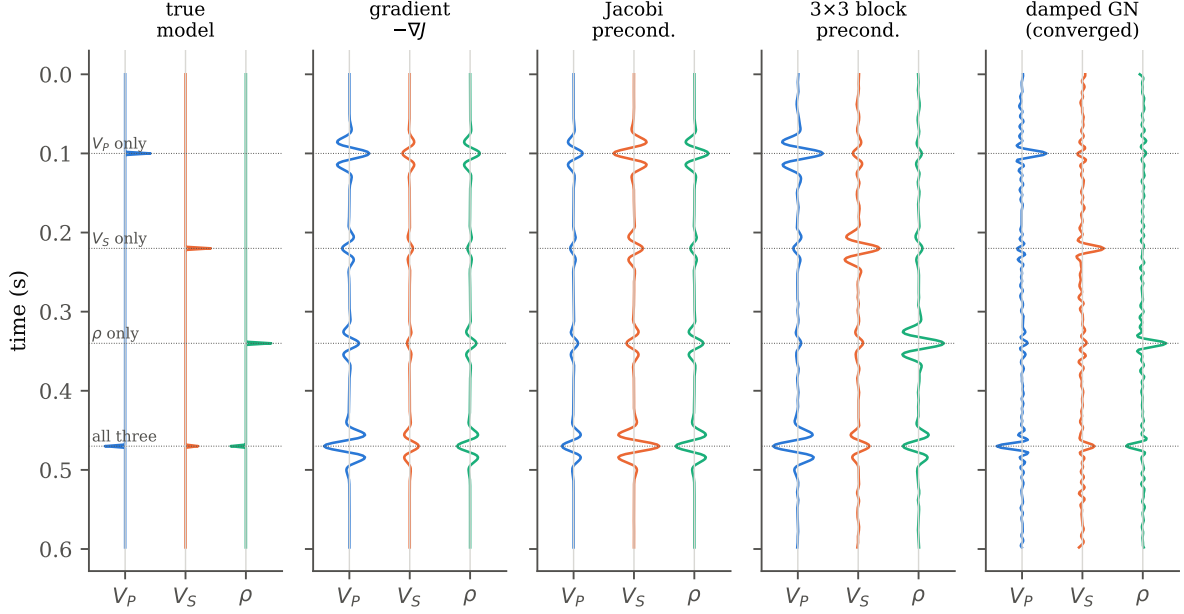


Figure 7: Laboratory traces ( $V_P$  blue,  $V_S$  orange,  $\rho$  green; each panel normalised). From left to right: true scatterers; the raw gradient  $-\nabla J = \mathbf{L}^T \mathbf{d}$  (CTI 0.41); the Jacobi-preconditioned gradient (CTI 0.75); the damped  $3 \times 3$  block-preconditioned gradient (CTI 0.06); the converged damped Gauss–Newton solution  $\delta \mathbf{m}^*$  (CTI 0.07). Dotted lines mark the scatterers. In the gradient, the  $\rho$ -only scatterer at 0.34s is as strong in  $V_P$  as in  $\rho$ , and the  $V_P$ -only scatterer at 0.10s leaks strongly into density. The block preconditioner assigns each event to the correct parameter after a single application.

- **Steepest descent** reduces cross-talk only slowly (CTI 0.41  $\rightarrow$  0.28 after 40 iterations), and convergence stalls at 59% distance. The zig-zag along the impedance trough of Figure 6 is the reason.
- **CG** (Gauss–Newton/truncated Newton) is faster and keeps improving (CTI 0.17 and distance 0.36 at 40 iterations), but its early iterates are almost as contaminated as the gradient. The cross-talk is removed only as CG builds up curvature information.
- **Jacobi preconditioning is counter-productive** for coupling: it worsens both measures throughout.
- **Block preconditioning** gives the best of both. Applied once, it removes most of the cross-talk (CTI 0.06). As a CG preconditioner it converges fastest (distance 0.17 at 40 iterations), because the remaining work is the spatial (wavelet) deconvolution, which CG handles efficiently. The small CTI rise over iterations, towards the  $\delta \mathbf{m}^*$  value 0.07, is the fitting of noise and wavelet side lobes, i.e. trade-off rather than coupling.

#### 9.4 The limits: aperture, noise and damping

Finally, Table 7 lists the CTI of the *converged* damped Gauss–Newton solution for different apertures, noise levels and damping strengths. This is the cross-talk that *no* optimiser can remove: trade-off.

The table summarises this tutorial in eight rows:

- With wide aperture and clean data, Gauss–Newton removes cross-talk almost completely ( $0.32 \rightarrow 0.01$  at  $60^\circ$ ). Coupling is curable.

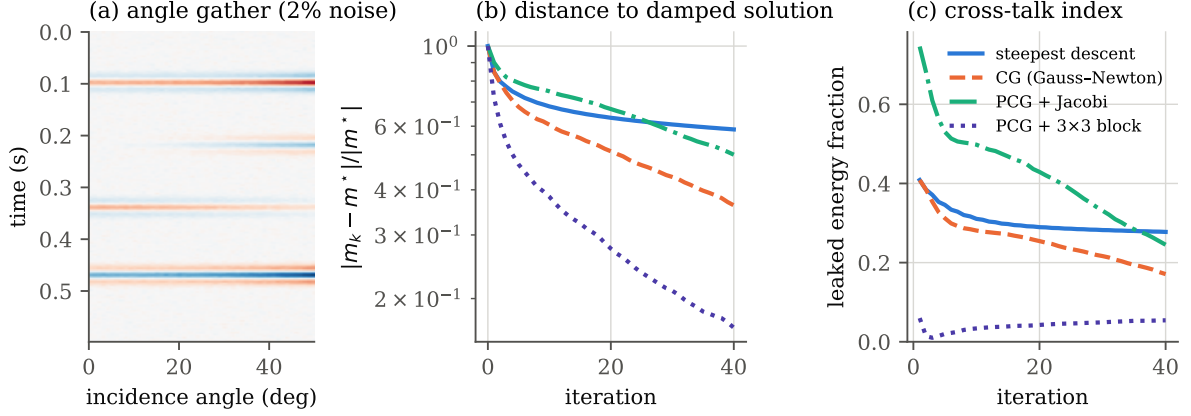


Figure 8: (a) The noisy angle gather (0–50°). (b) Relative distance to the damped least-squares solution  $\delta \mathbf{m}^*$  versus iteration, for steepest descent with exact line search, CG (i.e. Gauss–Newton solved by CG, the inner loop of truncated Newton), CG with Jacobi preconditioning and CG with  $3 \times 3$  block preconditioning. (c) Cross-talk index of the iterate.

Table 6: Laboratory convergence: distance to  $\delta \mathbf{m}^*$  / cross-talk index after  $k$  iterations ( $i_{\max} = 50^\circ$ , 2% noise).

| method                   | $k = 1$   | $k = 3$   | $k = 5$   | $k = 10$  | $k = 20$  | $k = 40$  |
|--------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| steepest descent         | .85 / .41 | .77 / .37 | .73 / .35 | .68 / .31 | .63 / .29 | .59 / .28 |
| SD + Jacobi              | .90 / .75 | .84 / .65 | .81 / .60 | .78 / .54 | .76 / .51 | .73 / .49 |
| SD + $3 \times 3$ block  | .72 / .06 | .58 / .02 | .53 / .02 | .46 / .02 | .41 / .03 | .36 / .03 |
| CG (GN / TN inner)       | .85 / .41 | .74 / .35 | .68 / .31 | .61 / .28 | .51 / .25 | .36 / .17 |
| PCG + Jacobi             | .90 / .75 | .82 / .61 | .79 / .53 | .75 / .50 | .67 / .43 | .50 / .25 |
| PCG + $3 \times 3$ block | .72 / .06 | .54 / .01 | .47 / .02 | .38 / .03 | .28 / .04 | .17 / .05 |

- With narrow aperture (30°) the inverse Hessian cannot do much (best 0.19 even without noise). With noise the weakly determined direction is either amplified (small  $\mu$ : 0.44) or suppressed by damping, which leaves residual leakage (0.34). That is trade-off: only more data (aperture, P–S) or prior information can help.
- The optimal damping grows with the noise level. With noise, the CTI is minimised at intermediate  $\mu$  ( $10^{-3}$ ), illustrating the bias–variance trade-off of regularisation.

## 10 Implementation guidance

### 10.1 Verification tests

Multiparameter codes have  $N_p$  times more places to hide a sign or scale error, and cross-talk symptoms are easily mistaken for bugs, and bugs for cross-talk. Run these tests on a small model before every production campaign:

1. **Gradient (Taylor) test, per class.** For a random smooth  $\mathbf{v}$  confined to class  $p$ , compare  $(J(\mathbf{m} + h\mathbf{v}) - J(\mathbf{m} - h\mathbf{v}))/2h$  with  $\mathbf{g}^T \mathbf{v}$  for  $h$  over several decades. The relative error should fall as  $h^2$  until round-off takes over. Repeat for every class: a correct  $V_P$  gradient does not imply a correct  $\rho$  gradient. (The 2-D reference in `tdfwi-hess/reference` reports a 2.7% agreement, which is adequate for a coarse control-point parameterization.)
2. **Dot-product (adjoint) test** of the Born operator, if one exists:  $\langle \mathbf{L}\delta\mathbf{m}, \mathbf{d} \rangle = \langle \delta\mathbf{m}, \mathbf{L}^\dagger \mathbf{d} \rangle$

Table 7: Cross-talk index of the raw gradient and of the converged damped Gauss–Newton solution  $(\mathbf{H}_{\text{GN}} + \mu \mathbf{I})^{-1} \mathbf{L}^T \mathbf{d}$ , with  $\mu$  given relative to  $\max \text{diag } \mathbf{H}_{\text{GN}}$ .

| $i_{\text{max}}$ | noise | gradient | $\mu = 10^{-4}$ | $\mu = 10^{-3}$ | $\mu = 10^{-2}$ |
|------------------|-------|----------|-----------------|-----------------|-----------------|
| 30°              | 0     | 0.49     | 0.19            | 0.30            | 0.33            |
| 30°              | 2%    | 0.49     | 0.44            | 0.36            | 0.34            |
| 40°              | 0     | 0.46     | 0.07            | 0.16            | 0.29            |
| 40°              | 2%    | 0.47     | 0.12            | 0.16            | 0.29            |
| 50°              | 0     | 0.41     | 0.03            | 0.06            | 0.16            |
| 50°              | 2%    | 0.41     | 0.16            | 0.08            | 0.17            |
| 60°              | 0     | 0.32     | 0.01            | 0.02            | 0.06            |
| 60°              | 2%    | 0.32     | 0.11            | 0.04            | 0.06            |

to near machine precision (the companion SymPy study obtains  $3 \times 10^{-15}$  in float64).

3. **Hessian symmetry test:**  $\langle \mathbf{v}_1, \mathbf{H} \mathbf{v}_2 \rangle = \langle \mathbf{v}_2, \mathbf{H} \mathbf{v}_1 \rangle$  for random  $\mathbf{v}_1, \mathbf{v}_2$  in *different* classes. This specifically checks the off-diagonal blocks. With finite-difference products, expect agreement to the level of the FD error, a few percent in float32. Larger asymmetry indicates an inconsistent gradient (optimise-then-discretise mismatch, non-differentiable post-processing such as clipping or muting, or a parameter-dependent mask).
4. **Hessian linearity:**  $\mathbf{H}(a\mathbf{v}) = a \mathbf{H} \mathbf{v}$  and  $\mathbf{H}(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{H} \mathbf{v}_1 + \mathbf{H} \mathbf{v}_2$ , again to FD accuracy. Failures expose a step size that is too large (nonlinearity) or too small (noise).
5. **Single-class recovery:** data generated with an anomaly in one class only, inverted for all classes, so that leakage can be measured (Section 7.6).

## 10.2 Choosing the finite-difference step

For the FD product (30), the step  $\epsilon$  multiplies a *multiparameter* direction  $\mathbf{v}$ . A seemingly natural choice,  $\epsilon = \epsilon_{\text{rel}} \|\mathbf{m}\|_2 / \|\mathbf{v}\|_2$  over the joint vector, fails when classes have different units. In one of this project’s test models,  $\|V_P\|_2 \approx 3.7 \times 10^6$  (m/s) and  $\|\rho\|_2 \approx 1.2 \times 10^3$  (g/cm<sup>3</sup>), so the joint norm is dominated by  $V_P$ . The resulting step perturbed density by  $\pm 6$  g/cm<sup>3</sup>, made it negative, and the propagator produced NaNs. The robust rule used in `tdfwi-hess` bounds the *relative* peak perturbation of every class:

$$\epsilon = \epsilon_{\text{rel}} \min_p \frac{\|\mathbf{m}_p\|_{\infty}}{\|\mathbf{v}_p\|_{\infty}}, \quad (52)$$

complemented by a physical clip of the perturbed model ( $V_P$  within bounds,  $V_S \geq 0$ ,  $\rho > 0$ ) as a safety net. With float32 gradients,  $\epsilon_{\text{rel}} \approx 10^{-4}$ – $10^{-3}$  for forward differences and  $\approx 10^{-3}$  for central differences (Figure 1a). A clean alternative is to work entirely in normalised (logarithmic) inversion parameters, where all classes are  $\mathcal{O}(1)$  and a single scalar step is natural.

### Common pitfalls with FD Hessian-vector products

- *Using the post-processed gradient.* Illumination division, smoothing, depth gain and masks are part of the *preconditioner*, not of  $\mathbf{H}$ . The FD difference must use the raw gradient  $\mathbf{L}^T \mathbf{r}$ ; apply the preconditioner separately as  $\mathbf{M}^{-1}$ .
- *Inconsistent randomness.* Random source encoding or shot subsampling must use the *same* realisation for  $\mathbf{g}(\mathbf{m})$  and  $\mathbf{g}(\mathbf{m} + \epsilon \mathbf{v})$ , or the difference is dominated by the change of subset.
- *Adaptive steps.* Re-estimating the source wavelet, the data normalisation or the mute between the two gradient evaluations changes the objective function. Freeze them

during the inner loop.

- *Non-smooth operations.* Model clipping,  $V_S$  masks in water and gradient muting are non-differentiable. Keep  $\epsilon$  small enough to stay inside their smooth regions.

### 10.3 Cost accounting

Let  $C$  be the cost of one wave simulation for all sources. Table 8 lists the costs of the building blocks.

Table 8: Cost of the building blocks per outer iteration ( $C =$  one simulation for all sources).

| operation                                  | cost                                      | memory                                                |
|--------------------------------------------|-------------------------------------------|-------------------------------------------------------|
| objective                                  | $C$                                       | —                                                     |
| gradient (all $N_p$ classes)               | $2C$                                      | forward strains/acceleration (storage or checkpoints) |
| line search (per trial)                    | $C$                                       | —                                                     |
| GN Hessian-vector (Born + adjoint)         | $2C$                                      | background field                                      |
| full Hessian-vector (second-order adjoint) | $2C$                                      | background and adjoint fields                         |
| FD Hessian-vector, forward / central       | $2C / 4C$                                 | as for the gradient                                   |
| NLCG / l-BFGS iteration                    | $\approx (2 + 1.5)C$                      | $\ell$ model-size vectors (l-BFGS)                    |
| truncated-Newton iteration                 | $\approx (2 + 2n_{cg} + 1.5)C$            | a few model-size vectors                              |
| block $N_p \times N_p$ preconditioner      | $\ll C$ (given $\mathbf{B}(\mathbf{x})$ ) | $N_p(N_p + 1)/2$ fields                               |

### 10.4 A recommended workflow

1. **Feasibility.** From the acquisition geometry and the background model, compute the scattering-angle coverage versus depth and evaluate the local kernels (47) for the candidate parameterizations (Table 2). Decide which classes are resolvable at which depths and wavenumbers, and which must be constrained.
2. **Kinematic stage.** Invert early arrivals and diving waves for  $V_P$  (or the dominant anisotropic velocity), from low to high frequencies. Hold the other classes fixed or tied by rock physics.
3. **Multiparameter stage.** Release the classes simultaneously in a reflection-friendly parameterization, with normalised variables, a class-wise diagonal illumination preconditioner and preferably a block preconditioner. Use l-BFGS, or truncated Newton with  $n_{cg} = 3-8$  when the early cross-talk is unacceptable.
4. **Constrain the trade-off.** Apply stronger smoothing or damping to density, and add structural coupling and rock-physics priors. Include P–S data if available.
5. **QC.** After each band, compare class-wise updates with single-class synthetic leakage tests, check  $V_P/V_S$  plausibility, and check that the data residual reduction is not driven by a single class absorbing modelling errors (Section 7.5).

## 11 Mapping the theory onto `tdfwi-hess`

The `tdfwi-hess` directory adds a matrix-free Hessian and truncated-Newton capability to the production multiparameter elastic FWI, *without modifying* the production `tdfwi` package (by subclassing). Table 9 lists where each concept of this tutorial lives.

Table 9: Concept-to-code map (paths relative to the repository root).

| Concept                                                | Implementation                                                                                                                                      |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Gradient $\mathbf{L}^\dagger \mathbf{r}$ , all classes | <code>rtmop()</code> in <code>tdfwi/pyrtm.cu</code> (mode <code>tdfwimod=0</code> ); raw per-class gradient <code>result['img']</code>              |
| Elastic imaging conditions (26)                        | <code>tdfwi/image_condition_mp_efwi.cu</code> ; symbolic proof of equivalence in <code>docs/SymPy_EFWI</code>                                       |
| Parameterizations (17)                                 | <code>-parameterization</code> (7 options); model parameters $(V_P, V_S, \rho)$ vs inversion parameters; <code>update_x</code>                      |
| Diagonal pseudo-Hessian (33)                           | source illumination <code>spower</code> (source dilatation energy); <code>tdfwi_obj_hess.illumination_precond()</code>                              |
| Raw gradient for Hessian work                          | <code>tdfwi_obj_hess._raw_grad()</code> : no illumination, smoothing or gain; mask only                                                             |
| FD Hessian-vector (30)                                 | <code>hessian.fd_hessian_vector</code> ; <code>tdfwi_obj_hess.hv_product()</code>                                                                   |
| FD step rule (52)                                      | <code>tdfwi_obj_hess._fd_step()</code> (per-class $\infty$ -norm) and the physical clip in <code>eval_at_steps()</code>                             |
| Truncated Newton (Algorithm 1)                         | <code>hessian.newton_cg</code> (relative negative-curvature test, forcing term, descent safeguard); <code>tdfwi_obj_hess.newton_search_dir()</code> |
| Gradient Taylor test                                   | <code>tdfwi_obj_hess.grad_qc()</code>                                                                                                               |
| Hessian symmetry QC                                    | <code>fwi_newton.py -hv_qc=true</code>                                                                                                              |
| Driver options                                         | <code>-optalg=NEWTONCG -newton_maxcg -newton_forcing -hv_eps_rel -hv_diff</code>                                                                    |
| CPU references                                         | <code>tests/test_hessian_cpu.py</code> ;<br><code>reference/mp_elastic_hessian_reference.py</code> ;<br><code>reference/demo_2d_mpefwi.py</code>    |

### 11.1 What the current implementation does, in the language of this tutorial

- **Hessian.** The *full* Hessian action is obtained by forward (default) or central finite differences of the raw gradient, at a cost of  $2C$  or  $4C$  per product (Section 5.3). Near convergence this approximates  $\mathbf{H}_{\text{GN}}$ . Far from it, it includes  $\mathbf{Q}$  and may show negative curvature, which `newton_cg` detects with a *relative* test.
- **Preconditioner.** The inner CG is preconditioned by the diagonal source-illumination pseudo-Hessian, with the *same* denominator  $(\text{spower} + \varepsilon)^{\text{spow\_exp}}$  applied to every class. This balances spatial illumination but not class-wise sensitivity, and it leaves all inter-parameter coupling to the CG iterations themselves (Section 8.4).
- **Cross-talk removal.** This is carried entirely by the Krylov approximation of  $\mathbf{H}^{-1}$  in `n_cg = -newton_maxcg` inner iterations. The 2-D reference demo reports a local inter-parameter correlation of 0.97–0.98 between  $V_P$  and  $\rho$  for its geometry. In that demo, five Newton-CG iterations reduced the objective  $3.6\times$  further than five steepest-descent iterations ( $1.3 \times 10^{-5}$  versus  $4.6 \times 10^{-5}$  of  $J_0 = 1.8 \times 10^{-3}$ ) and the relative model error from 0.93 to 0.84.

### 11.2 Natural next steps suggested by the theory

1. **Class-wise scaling.** Invert in normalised or logarithmic variables, or at least scale each class’s preconditioned gradient to comparable RMS (Section 8.2). This makes a single FD step and a single line-search step meaningful for all classes, and it simplifies (52).
2. **Class-specific diagonal.** Replace the shared `spower` denominator by per-class pseudo-Hessians: dilatation energy for  $V_P$ , strain-deviator energy for  $V_S$ , and kinetic-plus-strain energy for  $\rho$ . These follow directly from the virtual sources of (26).

3. **Block  $3 \times 3$  preconditioner.** Estimate  $\mathbf{B}(\mathbf{x})$  by Hessian probing with three sparse combs (one per class, 3 FD products =  $6C$ , amortised over many outer iterations if the background changes slowly), or from angle coverage and the radiation patterns. Apply its damped inverse (50) as the CG preconditioner. The laboratory suggests this is the single most effective addition (Table 6).
4. **Gauss–Newton product.** Add a hook in `rtmop` to inject a user-supplied adjoint source, so that  $\mathbf{L}^\dagger(\mathbf{L}\mathbf{v})$  can be formed from FD-linearised data  $\mathbf{L}\mathbf{v} \approx (\mathcal{F}(\mathbf{m} + \epsilon\mathbf{v}) - \mathcal{F}(\mathbf{m}))/\epsilon$ , which the existing `eval_alpha` line search already computes. This guarantees an SPD inner system.
5. **Leakage QC.** Add a single-class synthetic test to the regression suite (Section 10.1) to track the cross-talk index of each release.

## 12 Summary

### The ten things to remember

1. The gradient is the true model blurred by the Hessian:  $-\mathbf{g} = \mathbf{H}_{\text{GN}}\delta\mathbf{m}_{\text{true}}$ . Its off-diagonal blocks  $\mathbf{L}_p^\dagger\mathbf{L}_q$  are the cross-talk.
2. Locally, each Hessian block is an angle integral of products of radiation patterns (47). Cross-talk is the overlap of those patterns over the available aperture.
3. P–P radiation patterns at reflection angles *are* the Aki–Richards AVO coefficients (43). Short-offset P–P data constrain about two combinations of three elastic parameters.
4.  $V_P$  is a monopole.  $V_S$  is invisible in P–P transmission and backscattering. Density duplicates  $V_P$  in backscattering (velocity basis) or vanishes there (impedance basis).
5. Scattering angle maps to model wavenumber. Parameters invisible in transmission cannot be recovered at long wavelengths from P–P data.
6. Distinguish *imbalance* (fixed by scaling), *coupling* (fixed by approximating  $\mathbf{H}^{-1}$ ) and *trade-off* (fixed only by more data or prior information), and keep *modelling-error leakage* separate from all three.
7. Choose the parameterization for the data regime: impedances for reflections, velocities for transmissions. Always normalise.
8. Diagonal preconditioning balances illumination but can *amplify* coupling. Block  $N_p \times N_p$  preconditioning removes local coupling cheaply, and it must be damped.
9. Truncated Newton with Hessian-vector products (second-order adjoint or finite difference) removes coupling globally, at  $2n_{\text{cg}}$  extra simulations per iteration. Precondition it, test curvature *relatively*, and scale FD steps per class.
10. Aperture and wave-mode diversity (long offsets, converted waves) beat every algorithm. Assess them first with a radiation-pattern analysis.

**Further reading.** Virieux and Operto (2009) and Fichtner (2010) provide general background. Operto et al. (2013) is a practical multiparameter tour. Métivier et al. (2017) covers truncated Newton. Plessix (2006) reviews the adjoint state and Fichtner and Trampert (2011a) derives Hessian kernels. Köhn et al. (2012) and Pan et al. (2018) analyse elastic parameterizations, and Gholami et al. (2013) and Alkhalifah and Plessix (2014) anisotropic ones. Tarantola (2005) is

the reference on the probabilistic view.

## A Notation

| symbol                             | meaning                                                      | symbol                                           | meaning                                             |
|------------------------------------|--------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| $\mathbf{m}, \delta\mathbf{m}$     | model, perturbation                                          | $N_p, N_x$                                       | number of parameter classes, grid points            |
| $\mathcal{A}(\mathbf{m})$          | wave operator                                                | $\mathcal{A}_p$                                  | $\partial\mathcal{A}/\partial m_p$                  |
| $\mathbf{u}, \boldsymbol{\lambda}$ | forward, adjoint wavefield                                   | $\delta\mathbf{u}, \delta\boldsymbol{\lambda}$   | first-, second-order adjoint fields                 |
| $\mathcal{F}, \mathcal{R}$         | forward map, receiver sampling                               | $\mathbf{L}, \mathbf{L}_p$                       | Jacobian (Born), its class- $p$ block               |
| $\mathbf{r}$                       | residual $\mathcal{F}(\mathbf{m}) - \mathbf{d}^{\text{obs}}$ | $J$                                              | misfit                                              |
| $\mathbf{g}$                       | gradient $\mathbf{L}^\dagger \mathbf{r}$                     | $\mathbf{H}, \mathbf{H}_{\text{GN}}, \mathbf{Q}$ | full, Gauss–Newton, second-order Hessian            |
| $\mathbf{B}(\mathbf{x})$           | point-wise $N_p \times N_p$ block                            | $C_{pq}$                                         | correlation coefficient (48)                        |
| $\kappa, \kappa_J$                 | condition number, after Jacobi scaling                       | $\mathbf{T}$                                     | change-of-variables Jacobian                        |
| $V_P, V_S, \rho$                   | P, S velocity, density                                       | $I_P, I_S$                                       | P, S impedance                                      |
| $\lambda, \mu$                     | Lamé parameters                                              | $\gamma$                                         | background $V_S/V_P$                                |
| $\epsilon, \delta$                 | Thomsen parameters                                           | $\theta, i$                                      | scattering, incidence angle ( $\theta = \pi - 2i$ ) |
| $\mathcal{S}_p(\theta)$            | radiation pattern of class $p$                               | $\mathbf{k}$                                     | scattering wavenumber                               |
| $\epsilon, \epsilon_{\text{rel}}$  | FD step, relative FD step                                    | $n_{\text{cg}}, \eta_k$                          | inner CG iterations, forcing term                   |

## B Derivation of the Born scattering amplitude

Consider the frequency-domain elastic equation  $-\omega^2 \rho u_i - \partial_j (c_{ijkl} \partial_l u_k) = f_i$  in a homogeneous background  $(\rho, c_{ijkl})$ , perturbed by  $(\delta\rho, \delta c_{ijkl})$  localised near the origin. From (34), the scattered field satisfies the background equation with body force

$$\delta f_i = \omega^2 \delta\rho u_i^0 + \partial_j (\delta c_{ijkl} \partial_l u_k^0). \quad (53)$$

Take an incident plane wave  $\mathbf{u}^0 = \mathbf{a} e^{i\omega \mathbf{n} \cdot \mathbf{x} / c_{\text{in}}}$ , so that  $\partial_l u_k^0 = i(\omega/c_{\text{in}}) n_l a_k u^0$ . The far-field Green’s function for a wave of speed  $c_{\text{out}}$  leaving in direction  $\mathbf{q}$  with polarisation  $\mathbf{g}$  is  $G_{mi} \approx \frac{g_m g_i}{4\pi \rho c_{\text{out}}^2 r} e^{i\omega r/c_{\text{out}}} e^{-i\omega \mathbf{q} \cdot \mathbf{x} / c_{\text{out}}}$ . Projecting the scattered displacement  $\delta u_m = \int G_{mi} \delta f_i$  onto  $\mathbf{g}$  and integrating the stiffness term by parts moves the derivative  $\partial_j$  onto the Green’s function, which contributes  $-i(\omega/c_{\text{out}}) q_j$ . The amplitude of a point perturbation is then

$$\mathbf{g} \cdot \delta \mathbf{u} \propto \omega^2 \left[ \delta\rho (\mathbf{g} \cdot \mathbf{a}) - \frac{1}{c_{\text{in}} c_{\text{out}}} q_j g_i \delta c_{ijkl} n_l a_k \right], \quad (54)$$

which, divided by  $\rho\omega^2$ , is (35). Substituting the isotropic  $\delta c_{ijkl}$ :

$$q_j g_i \delta c_{ijkl} n_l a_k = \delta\lambda (\mathbf{g} \cdot \mathbf{q})(\mathbf{n} \cdot \mathbf{a}) + \delta\mu [(\mathbf{g} \cdot \mathbf{n})(\mathbf{q} \cdot \mathbf{a}) + (\mathbf{g} \cdot \mathbf{a})(\mathbf{q} \cdot \mathbf{n})]. \quad (55)$$

$P$ – $P$  ( $\mathbf{a} = \mathbf{n}, \mathbf{g} = \mathbf{q}, c_{\text{in}} = c_{\text{out}} = V_P$ ):  $\mathcal{S} = \cos\theta \delta\rho/\rho - (\delta\lambda + 2\delta\mu \cos^2\theta)/(\rho V_P^2)$ , i.e. (41). With  $\delta\lambda = \delta\rho(V_P^2 - 2V_S^2) + \rho(2V_P\delta V_P - 4V_S\delta V_S)$  and  $\delta\mu = \delta\rho V_S^2 + 2\rho V_S\delta V_S$  this gives (39). With (18) it gives (40).  $P$ – $SV$  ( $\mathbf{a} = \mathbf{n}, \mathbf{g} \perp \mathbf{q}, \mathbf{g} \cdot \mathbf{n} = -\sin\theta, c_{\text{out}} = V_S$ ): the  $\delta\lambda$  term vanishes because  $\mathbf{g} \cdot \mathbf{q} = 0$ , and  $\mathcal{S} = -\sin\theta \delta\rho/\rho + \sin 2\theta \delta\mu/(\rho V_P V_S)$ . Since  $\delta\mu/(\rho V_P V_S) = \gamma(\delta\rho/\rho + 2\delta V_S/V_S)$ , this is (42).

## C The l-BFGS two-loop recursion

Given pairs  $(\mathbf{s}_j, \mathbf{y}_j)$ ,  $j = k - \ell, \dots, k - 1$ , with  $\rho_j = 1/(\mathbf{y}_j^\top \mathbf{s}_j)$ , the product  $\mathbf{d} = -\mathbf{B}_k^{-1} \mathbf{g}_k$  is formed without storing any matrix (Nocedal, 1980):

- 1:  $\mathbf{q} \leftarrow \mathbf{g}_k$
- 2: **for**  $j = k - 1, \dots, k - \ell$  **do**
- 3:    $\alpha_j \leftarrow \rho_j \mathbf{s}_j^\top \mathbf{q}; \quad \mathbf{q} \leftarrow \mathbf{q} - \alpha_j \mathbf{y}_j$

```

4: end for
5:  $\mathbf{z} \leftarrow \mathbf{B}_0^{-1} \mathbf{q}$  {insert the diagonal or block preconditioner here}
6: for  $j = k - \ell, \dots, k - 1$  do
7:    $\beta \leftarrow \rho_j \mathbf{y}_j^\top \mathbf{z}$ ;  $\mathbf{z} \leftarrow \mathbf{z} + (\alpha_j - \beta) \mathbf{s}_j$ 
8: end for
9: return  $\mathbf{d} = -\mathbf{z}$ 

```

The default  $\mathbf{B}_0^{-1} = \frac{\mathbf{s}_{k-1}^\top \mathbf{y}_{k-1}}{\mathbf{y}_{k-1}^\top \mathbf{y}_{k-1}} \mathbf{I}$  is a single scalar for all classes, which is why l-BFGS needs well-scaled parameters (Section 8.2). Using a class-wise or block preconditioner as  $\mathbf{B}_0^{-1}$  removes that weakness.

## D Exercises

1. Starting from (39), derive the Aki–Richards identity (43).
2. Show that, over  $[0, \pi]$  with uniform weight, the density pattern  $\cos \theta$  is orthogonal to both Lam  patterns in (41). Why does this not help surface reflection data?
3. For the acoustic  $(V_P, \rho)$  problem with reflections  $i \leq i_{\max}$ , compute the  $2 \times 2$  kernel analytically and find the weak eigenvector as  $i_{\max} \rightarrow 0$ . Interpret it physically.
4. Prove that a pointwise reparameterization cannot change the rank of  $\mathbf{H}_{\text{GN}}$  (use (17)). Give an example where it changes  $\kappa_J$  by  $10^4$  (Table 2).
5. Derive the second-order adjoint (28) by differentiating (20)–(22) in the direction  $\mathbf{v}$ .
6. With `make_figures.py`, change  $\gamma$  from 0.5 to 0.35 and repeat Table 2. Which couplings change most, and why?
7. In the laboratory, replace the  $3 \times 3$  block preconditioner by its *undamped* inverse with  $\mu = 0$  in the blocks only. Explain the behaviour of the cross-talk index with iteration.
8. Add P–S data to the laboratory forward operator using (42) transformed to reflection angles. Quantify the reduction of the CTI at  $i_{\max} = 30^\circ$ .

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